Binary Relations

Discrete Math, Fall 2025

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Relations

"In mathematics you don't understand things. You just get used to them."

— John von Neumann

René Descartes

Évariste Galois

Ernst Schröder

Michael Rabin

Herbert Wilf

Relations as Sets

Definition 1: A *binary relation* R on sets A and B is a subset of the Cartesian product $A \times B$.

Notation: If $R \subseteq A \times B$, we write "$a\ R\ b$" to mean that element $a \in A$ is *related* to element $b \in B$.

Formally, $a\ R\ b$ iff $\langle a, b \rangle \in R$.

Note: R is used to denote both the relation itself ($a\ R\ b$) *and* the set of pairs ($R \subseteq A \times B$).

Note: The *order* of elements in the pair *matters*: $\langle a, b \rangle \in R$ denotes that a is related to b, not the other way around, unless there is *another* pair $\langle b, a \rangle$ in the relation.

Example: $R = \{\langle n, k \rangle \mid n, k \in \mathbb{N} \text{ and } n < k\}$

Definition 2:

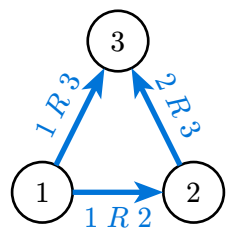
- A binary relation $R \subseteq A \times B$ on two different sets A and B is called *heterogeneous*.
- A binary relation $R \subseteq M^2$ on the same set M is called *homogeneous*.

Graph Representation

Definition 3: A homogeneous relation $R \subseteq M^2$ can be represented as a *directed graph* where:

- Vertices correspond to elements of M
- There is a directed edge from x to y if $x \; R \; y$, *i.e.* $\langle x, y \rangle \in R$

Example: For $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 1, 3 \rangle\}$, the graph has vertices $\{1, 2, 3\}$ and directed edges $1 \to 2$, $2 \to 3$, and $1 \to 3$.

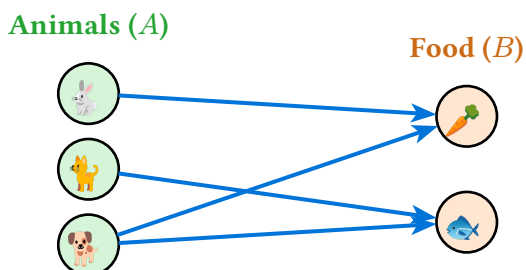


Graph Representation [2]

Definition 4: A heterogeneous relation $R \subseteq A \times B$ can be represented as a *bipartite graph* where:

- Vertices in one partition correspond to elements of A
- Vertices in the other partition correspond to elements of B
- There is a directed edge from $a \in A$ to $b \in B$ if $a \; R \; b$, *i.e.* $\langle a, b \rangle \in R$

Example: For animals $A = \{$ 🐰 , 🐈 , 🐕 $\}$, food $B = \{$ 🥕 , 🐟 $\}$, and relation R = "likes to eat", we have the bipartite graph with animal vertices on the left side and food vertices on the right side with four edges.



Matrix Representation

Definition 5: A binary relation $R \subseteq A \times B$ can be represented as a *matrix* $M_R = [\![R]\!]$ where:

- Rows correspond to elements of A
- Columns correspond to elements of B
- $M_R[i, j] = 1$ if $a_i \; R \; b_j$, and $M_R[i, j] = 0$ otherwise

Example: Let $A = \{a, b, c\}$, $B = \{x, y\}$, and $R = \{\langle a, x\rangle, \langle b, x\rangle, \langle c, y\rangle\}$. The matrix representation is:

$$[\![R]\!] = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{where rows are } \{a, b, c\} \text{ and columns are } \{x, y\}$$

Special Relations

Definition 6: For any set M, we define these special relations:

- *Empty relation*: $\emptyset \subseteq M^2$ (no elements are related)
- *Identity relation*: $I_M = \{\langle x, x\rangle \mid x \in M\}$ (each element related only to itself)
- *Universal relation*: $U_M = M^2$ (every element related to every element)

Example: For $M = \{a, b, c\}$:

- Empty: $\emptyset$
- Identity: $\{\langle a, a\rangle, \langle b, b\rangle, (c, c)\}$
- Universal: $\{\langle a, a\rangle, \langle a, b\rangle, \langle a, c\rangle \langle b, a\rangle, \langle b, b\rangle, \langle b, c\rangle, \langle c, a\rangle, \langle c, b\rangle, \langle c, c\rangle\}$ (all 9 pairs)

Operations on Relations

Definition 7: For relations $R, S \subseteq A \times B$:

- *Union*: $R \cup S = \{\langle a, b\rangle \mid \langle a, b\rangle \in R \text{ or } \langle a, b\rangle \in S\}$
- *Intersection*: $R \cap S = \{\langle a, b\rangle \mid \langle a, b\rangle \in R \text{ and } \langle a, b\rangle \in S\}$
- *Complement*: $\overline{R} = (A \times B) \setminus R$

Definition 8: For a relation $R \subseteq A \times B$, the *inverse* (or *converse*, or *dual*) relation is:

$$R^{-1} = \{\langle b, a\rangle \mid \langle a, b\rangle \in R\} \subseteq B \times A$$

Note: Other notations include R^T (transpose), R^C (converse), R° (reciprocal).

Example: If $R = \{\langle 1, x\rangle, \langle 2, y\rangle, \langle 2, z\rangle\}$, then $R^{-1} = \{\langle x, 1\rangle, \langle y, 2\rangle, \langle z, 2\rangle\}$.

Example: For the usual order relations, the converse is the naively "opposite" order:

- The converse of "less" is "greater": $<^T = >$
- The converse of "less or equal" is "greater or equal": $\leq^T = \geq$

Properties of Relations

Properties of Homogeneous Relations

Definition 9: A relation $R \subseteq M^2$ is *reflexive* if every element is related to itself:

$$\forall x \in M. (x \; R \; x)$$

Definition 10: A relation $R \subseteq M^2$ is *symmetric* if for every pair of elements, if one is related to the other, then the reverse is also true:

$$\forall x, y \in M. (x \; R \; y) \to (y \; R \; x)$$

Definition 11: A relation $R \subseteq M^2$ is *transitive* if for every three elements, if the first is related to the second, and the second is related to the third, then the first is also related to the third:

$$\forall x, y, z \in M. (x \; R \; y \land y \; R \; z) \to (x \; R \; z)$$

More Properties

Definition 12: A relation $R \subseteq M^2$ is *irreflexive* if no element is related to itself:

$$\forall x \in M. (x \not R\, x)$$

Definition 13: A relation $R \subseteq M^2$ is *antisymmetric* if for every pair of elements, if both are related to each other, then they must be equal:

$$\forall x, y \in M. (x\ R\ y \wedge y\ R\ x) \to (x = y)$$

Definition 14: A relation $R \subseteq M^2$ is *asymmetric* if for every pair of elements, if one is related to the other, then the reverse is not true:

$$\forall x, y \in M. (x\ R\ y) \to (y \not R\, x)$$

Note: *irreflexive* + *antisymmetric* = *asymmetric*.

Notes on Properties

- Reflexivity and irreflexivity are *not* mutually exclusive if $M = \emptyset$ (both are *vacuously*[1] `true`).
- Symmetry and antisymmetry are *not* mutually exclusive (*e.g.* identity relation).
- Asymmetry implies irreflexivity and antisymmetry.

[1] A statement "for all x in emptyset, $P(x)$" is considered `true` because there are *no counterexamples* in the empty set.

Additional Properties

Definition 15: A relation $R \subseteq M^2$ is:

- *Coreflexive* if $R \subseteq I_M$ (only related to themselves, if at all):

$$\forall x, y \in M. (x \; R \; y) \to (x = y)$$

- *Right Euclidean* if whenever an element is related to two others, those two are related:

$$\forall x, y, z \in M. (x \; R \; y \land x \; R \; z) \to (y \; R \; z)$$

- *Left Euclidean* if whenever two elements are both related to a third, they are related to each other:

$$\forall x, y, z \in M. (y \; R \; x \land z \; R \; x) \to (y \; R \; z)$$

Examples:

- Identity relation I_M is coreflexive. Any subset of I_M is also coreflexive.
- Equality relation "$=$" is left and right Euclidean.
- "Being in the same equivalence class" is Euclidean in both directions.

Equivalence Relations

Equivalence Relations

Definition 16: A relation $R \subseteq M^2$ is an *equivalence relation* if it is reflexive, symmetric and transitive.

Example (Equality): The *identity relation* $I_M = \{\langle x, x\rangle \mid x \in M\}$ is an equivalence relation on any set M.

Note: The identity relation is just the common equality relation "$=$".

Verification:

- **Reflexive:** $x \; I_M \; x$ for all $x \in M$ (by definition of I_M) ✓
- **Symmetric:** If $x \; I_M \; y$, then $x = y$, thus $y = x$, so $y \; I_M \; x$ ✓
- **Transitive:** If $x \; I_M \; y$ and $y \; I_M \; z$, then $x = y = z$, so $x \; I_M \; z$ ✓

This is the "finest" possible equivalence relation — it distinguishes every element.

Equivalence Classes

Definition 17: Let $R \subseteq M^2$ be an equivalence relation on a set M. The *equivalence class* of an element $x \in M$ under R is the set of all elements related to x:

$$[x]_R = \{y \in M \mid x \; R \; y\}$$

Example (Equality): For the identity relation I_M on set $M = \{a, b, c\}$:

- $[a]_{I_M} = \{a\}$ (only a is equal to a)
- $[b]_{I_M} = \{b\}$ (only b is equal to b)
- $[c]_{I_M} = \{c\}$ (only c is equal to c)

Each element forms its own equivalence class under equality.

Examples of Equivalence Relations

Example (Modular arithmetic): For any positive integer n, *congruence modulo n* on $\mathbb{Z}$ is defined by:

$$a \equiv b \pmod{n} \quad \text{iff} \quad n \mid (b - a)$$

Verification:

- **Reflexive:** $a \equiv a \pmod{n}$ since $n \mid (a - a) \iff n \mid 0$ ✓
- **Symmetric:** If $a \equiv b \pmod{n}$, then $n \mid (b - a) \iff n \mid (a - b)$, thus $b \equiv a \pmod{n}$ ✓
- **Transitive:** If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $n \mid (b - a)$ and $n \mid (c - b)$, so $n \mid ((c - b) + (b - a)) \iff n \mid (c - a)$, thus $a \equiv c \pmod{n}$ ✓

Equivalence classes (remainders): Let $n = 5$ and $M = \{0, 1, \ldots, 9\}$.

- $[0]_{\equiv} = \{0, 5\} = \{x \in M \mid x \equiv 0 \pmod{5}\}$
- $[1]_{\equiv} = \{1, 6\} = \{x \in M \mid x \equiv 1 \pmod{5}\}$
- $[2]_{\equiv} = \{2, 7\} = \{x \in M \mid x \equiv 2 \pmod{5}\}$
- $[3]_{\equiv} = \{3, 8\} = \{x \in M \mid x \equiv 3 \pmod{5}\}$
- $[4]_{\equiv} = \{4, 9\} = \{x \in M \mid x \equiv 4 \pmod{5}\}$

Examples of Equivalence Relations [2]

Example (Same absolute value): On $M = \{-3, -2, -1, 0, 1, 2, 3\}$, define relation R by:

$$x \; R \; y \quad \text{iff} \quad |x| = |y|$$

Verification:

- **Reflexive:** $|x| = |x|$ for all x ✓
- **Symmetric:** If $|x| = |y|$, then $|y| = |x|$ ✓
- **Transitive:** If $|x| = |y|$ and $|y| = |z|$, then $|x| = |z|$ ✓

Equivalence classes:

- $[0]_R = \{0\}$
- $[1]_R = [-1]_R = \{-1, 1\}$
- $[2]_R = [-2]_R = \{-2, 2\}$
- $[3]_R = [-3]_R = \{-3, 3\}$

Each positive number is equivalent to its negative counterpart.

Examples of Equivalence Relations [3]

Example (Same string length): On the set of all finite strings Σ^* over alphabet Σ, define:

$$s_1 \; R \; s_2 \quad \text{iff} \quad |s_1| = |s_2|$$

where $|s|$ denotes the length of string s.

Verification:

- **Reflexive:** Every string has the same length as itself ✓
- **Symmetric:** If two strings have the same length, the relation is symmetric ✓
- **Transitive:** If $|s_1| = |s_2|$ and $|s_2| = |s_3|$, then $|s_1| = |s_3|$ ✓

Equivalence classes: $[s]_R = \{t \in \Sigma^* \mid |t| = |s|\}$

For example, over $\Sigma = \{a, b\}$:

- $[\varepsilon]_R = \{\varepsilon\}$ (empty string)
- $[\mathrm{a}]_R = \{\mathrm{a}, \mathrm{b}\}$ (all strings of length 1)
- $[\mathrm{ab}]_R = \{\mathrm{aa}, \mathrm{ab}, \mathrm{ba}, \mathrm{bb}\}$ (all strings of length 2)

Examples of Equivalence Relations [4]

Example (Living in the same city): Let P be the set of people, and define relation R by:

$$p_1 \; R \; p_2 \quad \text{iff} \quad p_1 \text{ and } p_2 \text{ live in the same city}$$

Verification:

- **Reflexive:** Every person lives in the same city as themselves ✓
- **Symmetric:** If person A and B live in the same city, then B and A live in the same city ✓
- **Transitive:** If A and B live in the same city, and B and C do so, then A and C live in the same city ✓

Equivalence classes: Each equivalence class consists of all people living in the same city.

- $[\text{Alice}]_R$ = all people living in Alice's city
- This naturally partitions the population by cities

Application: This relation captures a common social grouping based on location.

Examples of Equivalence Relations [5]

Example (Similarity of triangles): Let T be the set of all triangles in the plane. Define relation $\sim$ by:

$$\triangle_1 \sim \triangle_2 \quad \text{iff} \quad \triangle_1 \text{ and } \triangle_2 \text{ are similar}^2$$

Verification:

- **Reflexive:** Every triangle is similar to itself ✓
- **Symmetric:** If triangle A is similar to triangle B, then triangle B is similar to triangle A ✓
- **Transitive:** If $A \sim B$ and $B \sim C$, then $A \sim C$ (similarity is transitive) ✓

Equivalence classes: Each class consists of all triangles with the same shape (but possibly different sizes).

- All equilateral triangles form one equivalence class
- All right triangles with legs in ratio 3:4:5 form another equivalence class

Geometric significance: This relation captures the concept of "same shape, different size."

[2] Two triangles are similar if their corresponding angles are equal.

Examples of Equivalence Relations [6]

Example (Rational numbers): On the set of ordered pairs $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, define:

$$\langle a, b\rangle \sim \langle c, d\rangle \quad \text{iff} \quad a \cdot d = b \cdot c$$

This relation is used to construct rational numbers $\mathbb{Q}$ from integer pairs.

Verification:

- **Reflexive:** $\langle a, b\rangle \sim \langle a, b\rangle$ since $a \cdot b = b \cdot a$ ✓
- **Symmetric:** If $a \cdot d = b \cdot c$, then $c \cdot b = d \cdot a$ ✓
- **Transitive:** If $a \cdot d = b \cdot c$ and $c \cdot f = d \cdot e$, then $a \cdot f = b \cdot e$ ✓

Equivalence classes: Each equivalence class represents a rational number.

- $[\langle 1, 2\rangle]_\sim = \{\langle 1, 2\rangle, \langle 2, 4\rangle, \langle 3, 6\rangle, \langle -1, -2\rangle, \ldots\}$ represents $\frac{1}{2}$
- $[\langle 3, 4\rangle]_\sim = \{\langle 3, 4\rangle, \langle 6, 8\rangle, \langle -3, -4\rangle, \ldots\}$ represents $\frac{3}{4}$

Mathematical significance: This construction shows how equivalence relations create new mathematical objects (rationals) from simpler ones (integers).

Quotient Sets

Definition 18: The *quotient set* of M by the equivalence relation R is the set of all equivalence classes:

$$M/_R = \{[x]_R \mid x \in M\}$$

Example: Consider $M = \{0, 1, 2, 3, 4, 5\}$ with the equivalence relation "congruent modulo 3":

$$x \equiv y \pmod 3 \quad \text{iff} \quad 3 \mid (x - y)$$

Equivalence classes:

- $[0]_\equiv = \{0, 3\} = \{x \in M \mid x \equiv 0 \pmod 3\}$
- $[1]_\equiv = \{1, 4\} = \{x \in M \mid x \equiv 1 \pmod 3\}$
- $[2]_\equiv = \{2, 5\} = \{x \in M \mid x \equiv 2 \pmod 3\}$

Quotient set:

$$M/_\equiv = \{\{0, 3\}, \{1, 4\}, \{2, 5\}\}$$

Note that $[3]_\equiv = [0]_\equiv = \{0, 3\}$, so we don't get a new equivalence class, since $M/_R$ is a *set*.

Quotient Sets [2]

Example: Let $M = \{\mathrm{a}, \mathrm{ab}, \mathrm{abc}, \mathrm{x}, \mathrm{xy}, \mathrm{z}\}$ with equivalence relation R defined by:

$$s_1 \; R \; s_2 \quad \text{iff} \quad |s_1| = |s_2|$$

where $|s|$ denotes the length of string s.

Equivalence classes by length:

- $[\mathrm{a}]_R = \{\mathrm{a}, \mathrm{x}, \mathrm{z}\}$ (strings of length 1)
- $[\mathrm{ab}]_R = \{\mathrm{ab}, \mathrm{xy}\}$ (strings of length 2)
- $[\mathrm{abc}]_R = \{\mathrm{abc}\}$ (strings of length 3)

Quotient set:

$$M/_R = \{\{\mathrm{a}, \mathrm{x}, \mathrm{z}\}, \{\mathrm{ab}, \mathrm{xy}\}, \{\mathrm{abc}\}\}$$

This partitions strings by their length, creating 3 equivalence classes.

Quotient Sets [3]

Example (Construction of rational numbers): Consider the set of ordered pairs $M = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$.

Define the equivalence relation $\sim$ by:

$$\langle a, b \rangle \sim \langle c, d \rangle \quad \text{iff} \quad a \cdot d = b \cdot c$$

Equivalence classes: Each represents a rational number in its "reduced form".

Representative	Equivalence Class	Rational Number
$\langle 1, 2 \rangle$	$\{\langle 1, 2 \rangle, \langle 2, 4 \rangle, \langle 3, 6 \rangle, \langle -1, -2 \rangle, \langle -2, -4 \rangle, \ldots\}$	$1/2$
$\langle 0, 1 \rangle$	$\{\langle 0, 1 \rangle, \langle 0, 2 \rangle, \langle 0, -3 \rangle, \langle 0, 7 \rangle, \ldots\}$	0
$\langle 3, 4 \rangle$	$\{\langle 3, 4 \rangle, \langle 6, 8 \rangle, \langle -3, -4 \rangle, \langle 9, 12 \rangle, \ldots\}$	$3/4$
$\langle -5, 3 \rangle$	$\{\langle -5, 3 \rangle, \langle 5, -3 \rangle, \langle -10, 6 \rangle, \langle 10, -6 \rangle, \ldots\}$	$-5/3$

Quotient set: All equivalence classes together form the set of rational numbers:

$$\mathbb{Q} := (\mathbb{Z} \times (\mathbb{Z} \setminus \{0\}))/_{\sim} = \{[\langle a, b \rangle]_{\sim} \mid \langle a, b \rangle \in \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})\}$$

Quotient Sets [4]

Interpretation: Each equivalence class $[\langle a, b\rangle]_\sim$ corresponds exactly to the rational number $\frac{a}{b}$.

- Different representations (a, b) and (c, d) belong to the same equivalence class iff they represent the same fraction: $\frac{a}{b} = \frac{c}{d}$
- The condition $a \cdot d = b \cdot c$ is the cross-multiplication test for fraction equality

Operations on the quotient set: We can define arithmetic operations on $\mathbb{Q}$ by:

$$[\langle a, b\rangle]_\sim + [\langle c, d\rangle]_\sim := [\langle ad + bc, bd\rangle]_\sim$$
$$[\langle a, b\rangle]_\sim \cdot [\langle c, d\rangle]_\sim := [\langle ac, bd\rangle]_\sim$$

These operations are *well-defined* because the result doesn't depend on the choice of representatives.

Mathematical significance:

- This construction shows that $\mathbb{Q}$ is literally *the* quotient set $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})/_\sim$
- It demonstrates how equivalence relations and quotient sets create new mathematical structures
- This is the rigorous foundation for rational number arithmetic taught in elementary school

Quotient Sets [5]

Example: Consider the set of all points in the plane $M = \mathbb{R}^2$ with the equivalence relation:

$$\langle x_1, y_1 \rangle \sim \langle x_2, y_2 \rangle \quad \text{iff} \quad x_1 + y_1 = x_2 + y_2$$

This relation groups points that lie on the same line of the form $x + y = c$ for some constant c.

Equivalence classes:

- $[\langle 0, 0 \rangle]_\sim = \{\langle x, y \rangle \mid x + y = 0\}$ (the line $x + y = 0$)
- $[\langle 1, 0 \rangle]_\sim = \{\langle x, y \rangle \mid x + y = 1\}$ (the line $x + y = 1$)
- $[\langle 0, 2 \rangle]_\sim = \{\langle x, y \rangle \mid x + y = 2\}$ (the line $x + y = 2$)

Quotient set:

$$\mathbb{R}^2/_\sim = \{L_c \mid c \in \mathbb{R}\}$$

where $L_c = \{\langle x, y \rangle \mid x + y = c\}$ is the line with equation $x + y = c$.

Each equivalence class is an entire line, and the quotient set consists of all such parallel lines.

Quotient Sets [6]

Example: Let $M = \{\text{apple}, \text{ant}, \text{banana}, \text{bee}, \text{cherry}, \text{cat}\}$ with equivalence relation:

$$w_1 \; R \; w_2 \quad \text{iff} \quad \text{first letter of } w_1 = \text{first letter of } w_2$$

Equivalence classes:

- $[\text{apple}]_R = \{\text{apple}, \text{ant}\}$ (words starting with 'a')
- $[\text{banana}]_R = \{\text{banana}, \text{bee}\}$ (words starting with 'b')
- $[\text{cherry}]_R = \{\text{cherry}, \text{cat}\}$ (words starting with 'c')

Quotient set:

$$M/_R = \{\{\text{apple}, \text{ant}\}, \{\text{banana}, \text{bee}\}, \{\text{cherry}, \text{cat}\}\}$$

This creates a dictionary-like grouping by first letter.

Quotient Sets [7]

Example (Trivial quotient sets):

Identity relation: If $R = I_M$ (identity relation), then each element forms its own equivalence class:

$$M/_{I_M} = \{\{x\} \mid x \in M\}$$

The quotient set has the same cardinality as the original set.

Universal relation: If $R = M \times M$ (universal relation), then all elements are equivalent:

$$M/_{M \times M} = \{M\}$$

The quotient set contains exactly one equivalence class — the entire set M.

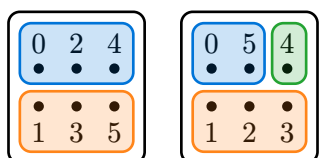
Set Partitions

Definition 19: A *partition* $\mathcal{P}$ of a set M is a family of non-empty, pairwise-disjoint subsets whose union is M:

- (Non-empty) $\forall B \in \mathcal{P}. (B \neq \emptyset)$
- (Disjoint) $\forall B_1, B_2 \in \mathcal{P}. (B_1 \neq B_2) \to (B_1 \cap B_2 = \emptyset)$
- (Cover) $\bigcup_{B \in \mathcal{P}} B = M$

Elements of $\mathcal{P}$ are *blocks* (or *cells*).

Example: For $M = \{0, 1, 2, 3, 4, 5\}$: $\{\{0, 2, 4\}, \{1, 3, 5\}\}$ and $\{\{0, 5\}, \{1, 2, 3\}, \{4\}\}$ are partitions.



Partitions and Equivalence Relations

Theorem 1 (Equivalences $\Leftrightarrow$ Partitions): Each equivalence relation R on M yields the partition $\mathcal{P}_R = \{[x]_R \mid x \in M\}$. Each partition $\mathcal{P}$ yields an equivalence $R_{\mathcal{P}}$ given by $\langle x, y\rangle \in R_{\mathcal{P}}$ iff x and y lie in the same block. These constructions invert one another.

Proof *(Sketch)*: Classes of an equivalence are non-empty, disjoint, and cover M. Conversely, "same block" relation is reflexive, symmetric, transitive. Composing the two constructions returns exactly the starting equivalence relation or partition (they are mutually inverse up to equality of sets of ordered pairs). $\square$

Composition of Relations

Composition of Relations

Definition 20: The *composition* of two relations $R \subseteq A \times B$ and $S \subseteq B \times C$ is defined as:

$$R \,;\, S = S \circ R = \{\langle a, c\rangle \mid \exists b \in B.\, (a \; R \; b) \wedge (b \; S \; c)\}$$

Example: Let $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$, $C = \{x, y\}$ with relations:

- $R = \{\langle 1, a\rangle, \langle 1, b\rangle, \langle 2, c\rangle, \langle 3, d\rangle\} \subseteq A \times B$
- $S = \{\langle a, x\rangle, \langle b, y\rangle, \langle c, x\rangle\} \subseteq B \times C$

To find $R \,;\, S$, we look for pairs (i, z) where there exists w such that $\langle i, w\rangle \in R$ and $\langle w, z\rangle \in S$:

From 1: can reach a and b via R

- a connects to x via $S \Rightarrow (1, x)$ is in the composition
- b connects to y via $S \Rightarrow (1, y)$ is in the composition

From 2: can reach c via R

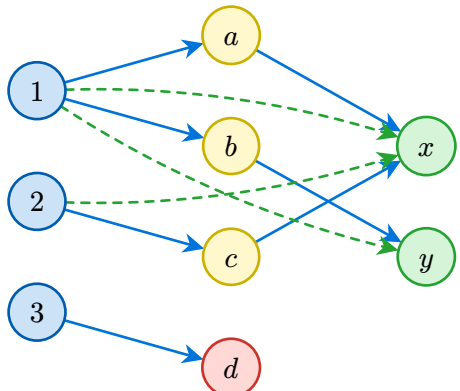
- c connects to x via $S \Rightarrow (2, x)$ is in the composition

From 3: can reach d via R

Composition of Relations [2]

- d has no outgoing connections in $S \Rightarrow$ no pairs from 3 in the composition

Therefore: $R \,;\, S = \{\langle 1, x\rangle, \langle 1, y\rangle, \langle 2, x\rangle\}$



Edges legend:

- Solid: Relations R and S
- Dashed: Composition $R \,;\, S$
- Red: Dead end (no outgoing path)

Examples of Composition

Example (Composition in a social network): Consider three sets:

- People $= \{\text{Alice}, \text{Bob}, \text{Carol}\}$
- Skills $= \{\text{Python}, \text{Design}, \text{Management}\}$
- Projects $= \{\text{WebApp}, \text{Mobile}, \text{Analytics}\}$

Define relations:

- HasSkill $= \{\langle \text{Alice}, \text{Python}\rangle, \langle \text{Alice}, \text{Design}\rangle, \langle \text{Bob}, \text{Python}\rangle, \langle \text{Carol}, \text{Management}\rangle\}$
- RequiresSkill $= \{\langle \text{Python}, \text{WebApp}\rangle, \langle \text{Python}, \text{Analytics}\rangle, \langle \text{Design}, \text{WebApp}\rangle, \langle \text{Management}, \text{Mobile}\rangle\}$

The composition HasSkill ; RequiresSkill gives us CanWorkOn:

- Alice can work on WebApp (via Python AND Design)
- Alice can work on Analytics (via Python)
- Bob can work on WebApp (via Python)
- Bob can work on Analytics (via Python)
- Carol can work on Mobile (via Management)

So: CanWorkOn $=$
$\{\langle \text{Alice}, \text{WebApp}\rangle, \langle \text{Alice}, \text{Analytics}\rangle, \langle \text{Bob}, \text{WebApp}\rangle, \langle \text{Bob}, \text{Analytics}\rangle, \langle \text{Carol}, \text{Mobile}\rangle\}$

Examples of Composition [2]

Example (Matrix composition): Relations can be composed using Boolean matrix multiplication.

Given $R \subseteq \{1, 2\} \times \{a, b\}$ and $S \subseteq \{a, b\} \times \{x, y\}$:

$$[\![R]\!] = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad [\![S]\!] = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The composition $[\![R \mathbin{;} S]\!] = [\![R]\!] \odot [\![S]\!]$ using Boolean matrix multiplication:

$$[\![R \mathbin{;} S]\!] = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) \\ (1 \wedge 1) \vee (1 \wedge 0) & (1 \wedge 1) \vee (1 \wedge 1) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

This means $R \mathbin{;} S = \{\langle 1, x\rangle, \langle 1, y\rangle, \langle 2, x\rangle, \langle 2, y\rangle\}$ (the universal relation).

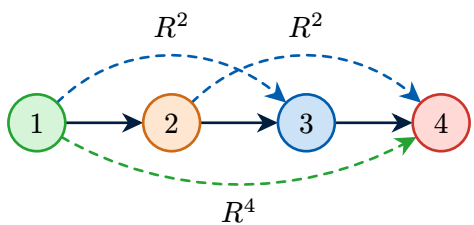
Powers of Relations

Definition 21: For a homogeneous relation $R \subseteq M^2$, we define *powers* of R:

- $R^0 = I_M$ (identity relation)
- $R^1 = R$
- $R^{n+1} = R^n \circ R$ for $n \geq 1$

Example: Let $M = \{1, 2, 3, 4\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 3, 4\rangle\}$ (successor relation).

- $R^1 = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 3, 4\rangle\}$
- $R^2 = \{\langle 1, 3\rangle, \langle 2, 4\rangle\}$ (two steps)
- $R^3 = \{\langle 1, 4\rangle\}$ (three steps)
- $R^4 = \emptyset$ (no four-step paths)

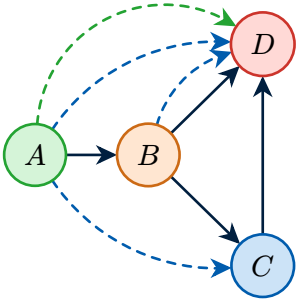


Paths in Graphs

Example (Path composition in a graph): Consider a directed graph with vertices $\{A, B, C, D\}$ and relation R representing direct edges: $R = \{\langle A, B\rangle, \langle B, C\rangle, \langle B, D\rangle, \langle C, D\rangle\}$

Powers of R represent paths of different lengths:

- $R^1 = R$ (direct connections)
- $R^2 = R \circ R$ (2-step paths):
  - $\langle A, C\rangle$: path $A \to B \to C$
  - $\langle A, D\rangle$: path $A \to B \to D$
  - $\langle B, D\rangle$: path $B \to C \to D$
  - So $R^2 = \{\langle A, C\rangle, \langle A, D\rangle, \langle B, D\rangle\}$.
- $R^3 = R^2 \circ R$ (3-step paths):
  - $\langle A, D\rangle$: path $A \to B \to C \to D$
  - No more 3-step paths!
  - So $R^3 = \{\langle A, D\rangle\}$.
- $R^4 = \emptyset$ (no 4-step paths)



Associativity of Composition

Theorem 2: Composition of relations is associative: $(R \mathbin{;} S) \mathbin{;} T = R \mathbin{;} (S \mathbin{;} T)$.

Proof: Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$ be three relations.

($\subseteq$): Let $\langle a, d\rangle \in (R \mathbin{;} S) \mathbin{;} T$.

- By definition of composition: $\exists c \in C. (\langle a, c\rangle \in R \mathbin{;} S) \wedge (\langle c, d\rangle \in T)$.
- Since $\langle a, c\rangle \in R \mathbin{;} S$, we have: $\exists b \in B. (\langle a, b\rangle \in R) \wedge (\langle b, c\rangle \in S)$.
- From $\langle b, c\rangle \in S$ and $\langle c, d\rangle \in T$, we have: $\langle b, d\rangle \in S \mathbin{;} T$.
- From $\langle a, b\rangle \in R$ and $\langle b, d\rangle \in S \mathbin{;} T$, we have: $\langle a, d\rangle \in R \mathbin{;} (S \mathbin{;} T)$.

($\supseteq$): Let $\langle a, d\rangle \in R \mathbin{;} (S \mathbin{;} T)$.

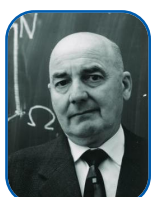
- By definition of composition: $\exists b \in B. (\langle a, b\rangle \in R) \wedge (\langle b, d\rangle \in S \mathbin{;} T)$.
- Since $\langle b, d\rangle \in S \mathbin{;} T$, we have: $\exists c \in C. (\langle b, c\rangle \in S) \wedge (\langle c, d\rangle \in T)$.
- From $\langle a, b\rangle \in R$ and $\langle b, c\rangle \in S$, we have: $\langle a, c\rangle \in R \mathbin{;} S$.
- From $\langle a, c\rangle \in R \mathbin{;} S$ and $\langle c, d\rangle \in T$, we have: $\langle a, d\rangle \in (R \mathbin{;} S) \mathbin{;} T$.

Therefore, $(R \mathbin{;} S) \mathbin{;} T = R \mathbin{;} (S \mathbin{;} T)$. $\square$

Order Theory

"Order is heaven's first law."

— Alexander Pope



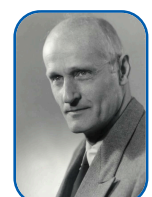
Helmut Hasse

Felix Hausdorff

Henri Poincaré

Robert Floyd

Stephen Warshall



Stephen Kleene

Orders

Definition 22: A relation $R \subseteq M^2$ is called a *preorder* if it is reflexive and transitive.

Definition 23: A *partial order* is a relation $R \subseteq M^2$ that is reflexive, antisymmetric, and transitive.

Definition 24: A relation $R \subseteq M^2$ is *connected* if for every pair of distinct elements, either one is related to the other or vice versa:

$$\forall x, y \in M.\, (x \neq y) \to (x \; R \; y \lor y \; R \; x)$$

Definition 25: A partial order which is also connected is called a *total order* (or *linear order*).

Examples of Orders

Example: The relation $\leq$ on real numbers $\mathbb{R}$ is a *total order*:

- **Reflexive:** $x \leq x$ for all $x \in \mathbb{R}$
- **Antisymmetric:** If $x \leq y$ and $y \leq x$, then $x = y$
- **Transitive:** If $x \leq y$ and $y \leq z$, then $x \leq z$
- **Connected:** For any $x, y \in \mathbb{R}$, either $x \leq y$ or $y \leq x$

This is the most familiar example of an order relation. Similarly, $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ with $\leq$ are all total orders.

Examples of Orders [2]

Example: The *subset relation* $\subseteq$ on the power set $\mathcal{P}(A)$ is a **partial order**.

Verification:

- **Reflexive:** $X \subseteq X$ for all $X \subseteq A$ ✓
  - Every set is a subset of itself by definition
- **Antisymmetric:** If $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$ ✓
  - If every element of X is in Y, and every element of Y is in X, then X and Y have the same elements
- **Transitive:** If $X \subseteq Y$ and $Y \subseteq Z$, then $X \subseteq Z$ ✓
  - If every element of X is in Y, and every element of Y is in Z, then every element of X is in Z

For $A = \{1, 2\}$, we have $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ with:

- $\emptyset \subseteq \{1\} \subseteq \{1, 2\}$ (this is a chain)
- $\emptyset \subseteq \{2\} \subseteq \{1, 2\}$ (another chain)
- But $\{1\}$ and $\{2\}$ are *incomparable* (neither is a subset of the other)

This is *not* a total order because not all pairs are comparable.

Examples of Orders [3]

Example: Consider the *no longer than* relation $\preccurlyeq$ on binary strings $\mathbb{B}^*$:

$$x \preccurlyeq y \quad \text{iff} \quad \text{len}(x) \leq \text{len}(y)$$

Verification:

- **Reflexive:** $x \preccurlyeq x$ for all $x \in \mathbb{B}^*$ ✓
  - $\text{len}(x) \leq \text{len}(x)$ is always true
- **Transitive:** If $x \preccurlyeq y$ and $y \preccurlyeq z$, then $x \preccurlyeq z$ ✓
  - If $\text{len}(x) \leq \text{len}(y)$ and $\text{len}(y) \leq \text{len}(z)$, then $\text{len}(x) \leq \text{len}(z)$ by transitivity of $\leq$
- **Antisymmetric:** ✗
  - Counter-example: $01 \preccurlyeq 10$ and $10 \preccurlyeq 01$ (since both have length 2), but $01 \neq 10$
- **Connected:** For any $x, y \in \mathbb{B}^*$, either $x \preccurlyeq y$ or $y \preccurlyeq x$ ✓
  - Either $\text{len}(x) \leq \text{len}(y)$ or $\text{len}(y) \leq \text{len}(x)$ (or both)

This is a *preorder* (reflexive and transitive), and even connected, but *not a partial order* due to lack of antisymmetry. Different strings of the same length are all "equivalent" under this relation, but they're not actually equal.

Examples of Orders [4]

Example: *Divisibility* $\mid$ on positive integers $\mathbb{N}^+$ is a **partial order**.

Verification:

- **Reflexive:** $n \mid n$ for all $n \in \mathbb{N}^+$ (every number divides itself) ✓
- **Antisymmetric:** If $a \mid b$ and $b \mid a$, then $a = b$ ✓
  - If a divides b, then $b = ak$ for some positive integer k
  - If b divides a, then $a = b\ell$ for some positive integer ℓ
  - Substituting: $b = ak = (b\ell)k = bk\ell$, so $k\ell = 1$
  - Since $k, \ell \in \mathbb{N}^+$, we must have $k = \ell = 1$, hence $a = b$
- **Transitive:** If $a \mid b$ and $b \mid c$, then $a \mid c$ ✓
  - If $a \mid b$, then $b = ak$ for some integer k
  - If $b \mid c$, then $c = b\ell$ for some integer ℓ
  - Therefore: $c = b\ell = (ak)\ell = a(k\ell)$, so $a \mid c$
- **Connected:** ✗
  - Counter-example: 2 does not divide 3, and 3 does not divide 2

This is a *partial order* but not a total order because some pairs are incomparable.

Examples of Orders [5]

Example: *Lexicographic order* on strings A^n (like dictionary order) is a **total order**.

Verification:

- **Reflexive:** $s \preceq s$ for all strings s ✓
  - A string is lexicographically equal to itself
- **Antisymmetric:** If $s \preceq t$ and $t \preceq s$, then $s = t$ ✓
  - If s comes before or equals t AND t comes before or equals s, then $s = t$
- **Transitive:** If $s \preceq t$ and $t \preceq u$, then $s \preceq u$ ✓
  - Lexicographic comparison preserves transitivity through character-by-character comparison
- **Connected:** For any strings s, t, either $s \preceq t$ or $t \preceq s$ ✓
  - We can always compare strings lexicographically by comparing character by character

For binary strings of length 2: $00 \prec 01 \prec 10 \prec 11$

Key property: Every pair of strings is *comparable*, making this a *total order*.

Partially Ordered Sets

Definition 26: A *partially ordered set* (or *poset*) $\langle S, \leq \rangle$ is a set S equipped with a partial order $\leq$.

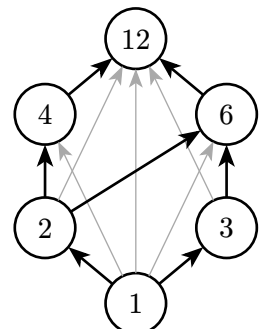
Example: Consider the poset $(D, |)$ where $D = \{1, 2, 3, 4, 6, 12\}$ and $|$ is divisibility.

Order relation properties:

- Reflexive: $n \mid n$ for all $n \in D$
- Antisymmetric: If $a \mid b$ and $b \mid a$, then $a = b$
- Transitive: If $a \mid b$ and $b \mid c$, then $a \mid c$

Special elements:

- Least element: 1 (divides all others)
- Greatest element: 12 (divisible by all others)
- Minimal elements: just 1
- Maximal elements: just 12



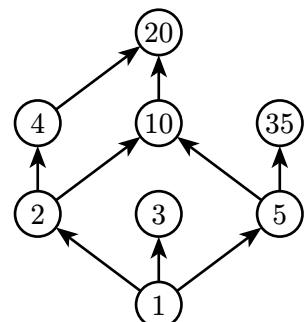
Note: This poset forms a *lattice* since every pair has a supremum (LCM) and infimum (GCD).

Hasse Diagrams

Definition 27: A *Hasse diagram* is a visual representation of a poset where:

- Each element is represented as a vertex.
- If $x < y$ and there is no z with $x < z < y$, draw an edge from x to y.
- Elements are arranged vertically by the order relation.
- Transitive connections are omitted (implied by transitivity of the partial order).

Example: For $D = \{1, 2, 3, 4, 5, 10, 20, 35\}$ with divisibility $|$.

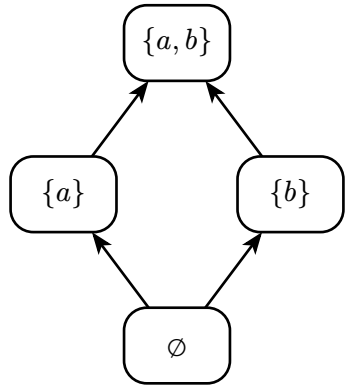


Reading the diagram:

- 1 divides everything (least element at bottom)
- Multiple *maximal* elements: 3, 20, 35 (no single greatest element)
- Some *chains* (not necessarily maximal, can skip elements):
  - Chain: $1 \mid 2 \mid 4$ (powers of 2)
  - Chain: $1 \mid 10 \mid 20$ (multiples of 10)
  - Chain: $5 \mid 35$ (multiples of 5)
- Primes 2, 3, 5 are *incomparable* to each other

Subset Poset

Example: For $\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ with inclusion $\subseteq$:



Note: This is the classic "diamond" shape characteristic of Boolean algebras.

Note: $\langle \mathcal{P}(A), \subseteq \rangle$ is also called the *Boolean lattice*.

Covering Relation

Definition 28: In a poset $\langle S, \leq \rangle$, an element $y \in S$ *covers* $x \in S$, denoted $x \lessdot y$, if there is no other element in between them:

$$x \lessdot y \quad \text{iff} \quad (x < y) \wedge \nexists z \in S. (x < z < y)$$

Note: "$<$" denotes the *induced strict order*:

$$x < y \quad \text{iff} \quad (x \leq y) \wedge (x \neq y)$$

Note: Hasse diagram is just a graph of a covering relation!

Maximal and Minimal Elements

Definition 29: An element $m \in S$ is called a *maximal element* of a poset $\langle S, \leq \rangle$ if it is not less than any other element, *i.e.*, there is no even greater element.

$$\forall x \neq m. \neg(x \leq m) \quad \Longleftrightarrow \quad \nexists x \neq m. (m \leq x)$$

Equivelently, $\forall x \in S. (m \leq x) \rightarrow (m = x)$

Definition 30: An element $m \in S$ is called a *minimal element* of a poset $\langle S, \leq \rangle$ if it is not greater than any other element, *i.e.*, there is no even smaller element.

$$\forall x \neq m. \neg(m \leq x) \quad \Longleftrightarrow \quad \nexists x \neq m. (x \leq m)$$

Equivelently, $\forall x \in S. (x \leq m) \rightarrow (m = x)$

Note: There may be multiple maximal (or minimal) elements.

Example of Maximal and Minimal Elements

Example: Consider the divisibility poset on $S = \{2, 3, 4, 6, 8, 12\}$:

Maximal elements: 8 and 12

- 8 divides nothing else in S except itself
- 12 divides nothing else in S except itself

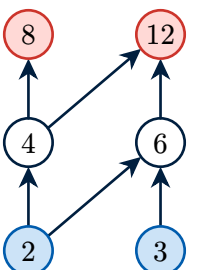
Minimal elements: 2 and 3

- Nothing in S properly divides 2 (since $1 \notin S$)
- Nothing in S properly divides 3

Note: This poset has no greatest or least element, but multiple minimal/maximal elements.

Hasse diagram:

- Maximal elements (8, 12) — they divide nothing else in S
- Minimal elements (2, 3) — nothing in S divides them
- Two separate chains: $2 \mid 4 \mid 8$ and $3 \mid 6 \mid 12$



Example of Maximal and Minimal Elements [2]

Example: Let $S = \{\text{ab}, \text{abc}, \text{abd}, \text{ac}, \text{b}, \text{bc}\}$ ordered by the *prefix* relation $\lessdot$:

$$x \lessdot y \text{ iff } x \text{ is a prefix of } y$$

Maximal elements: abc, abd, <u>ac</u>, bc

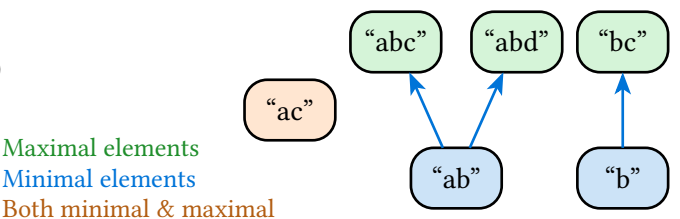
- These strings are not prefixes of any other string in S

Minimal elements: ab, <u>ac</u>, b

- These strings have no other string in S that is a proper prefix of them

The Hasse diagram shows three separate "trees" rooted at the minimal elements, with:

- $\text{ab} \lessdot \text{abc}$ and $\text{ab} \lessdot \text{abd}$
- $\text{b} \lessdot \text{bc}$
- ac stands alone (no other string extends it in S)



Greatest and Least Elements

Definition 31: The *greatest element* of a poset $\langle S, \leq \rangle$ is an element $g \in S$ that is greater than or equal to every other element in S, *i.e.*, for all $x \in S$, $x \leq g$.

Definition 32: A *least element* (bottom) b satisfies $b \leq x$ for all $x \in S$.

Note: Greatest (top) and least (bottom) elements are *unique* when they exist.

Examples:

- $\langle \mathcal{P}(A), \subseteq \rangle$: least $\emptyset$ (contained in every set), greatest A (contains every subset).
- $\langle \mathbb{N}^+, | \rangle$: least 1 (divides every positive integer), no greatest element (no integer is divisible by all others).
- $\langle \mathbb{Z}, \leq \rangle$: no least or greatest element (integers extend infinitely in both directions).
- $\langle \{1, 2, 3, 4, 5, 6\}, | \rangle$: least 1, no greatest element, maximal elements are 4, 5, 6 (prime powers and primes that don't divide anything else in the given set).

Converse Orders

Definition 33: The *dual* (or *converse*) of a poset $\langle S, \leq \rangle$ is the poset $\langle S, \geq \rangle$ where $x \geq y$ iff $y \leq x$.

Example: Consider the $\mathbb{N}^+$ ordered naturally.

- For $\leq$ order:
  - The *$\leq$-least* element is 1, since it is *$\leq$-smaller* than all others.
  - 1 is also $\leq$-minimal, since there are no other element which is $\leq$-smaller than 1.
  - There are *no $\leq$-maximal* elements, since the set is unbounded above.
  - On the Hasse diagram, 1 is at the bottom, and the diagram extends infinitely *upwards*.
- For $\geq$ order, minimal and maximal elements "flip":
  - There are *no $\geq$-minimal* elements. A $\geq$-minimal element would be an element m such that there is no other ($\geq$-smaller) element $n \neq m$ with $n \geq m$. However, for any $m \in \mathbb{N}^+$, there exists $n = m + 1$ which is $n \geq m$, so no such $\geq$-minimal element exists.
  - The *$\geq$-greatest* element is 1, since all elements are *$\geq$-smaller* than it.
  - On the Hasse diagram, 1 is at the top, and the diagram extends infinitely *downwards*.

Notes on Converse Orders

Note:

- Maximal elements in $\langle S, \leq \rangle$ become minimal in $\langle S, \geq \rangle$ and vice versa.
- Greatest element in $\langle S, \leq \rangle$ becomes least in $\langle S, \geq \rangle$ and vice versa.
- Chains and antichains remain the same in both orders.
- The Hasse diagram is flipped vertically when taking the dual order.
  - For $\langle \mathbb{N}^+, \leq \rangle$:
    - The *least* element 1 is at the bottom.
    - The diagram of $\langle \mathbb{N}^+, \leq \rangle$ extends infinitely *upwards*.
  - In the dual $\langle \mathbb{N}^+, \geq \rangle$:
    - 1 becomes the *greatest* element at the top.
    - The diagram of $\langle \mathbb{N}^+, \geq \rangle$ extends infinitely *downwards*.

Chains and Antichains

Definition 34: In a partially ordered set $(M, \leq)$:

- A *chain* is a subset $C \subseteq M$ where every two elements are comparable. Formally:

$$\forall x, y \in C. \, (x \leq y \text{ or } y \leq x)$$

- An *antichain* is a subset $A \subseteq M$ where no two distinct elements are comparable. Formally:

$$\forall x, y \in A. \, (x \neq y) \rightarrow (x \not\leq y \text{ and } y \not\leq x)$$

Note:

- Chains correspond to arbitrary paths or sub-sequences in the Hasse diagram.
- A *maximal chain* is a chain that cannot be extended by including any other elements from M.
- A *maximum chain* is a chain of the largest possible size in M.
- A chain is a *totally ordered subset* of the poset.
- An antichain consists of *pairwise incomparable elements*.
- Any singleton set is both a chain and an antichain.

Examples of Chains and Antichains

Example: Consider the divisibility poset $(D, |)$ where $D = \{1, 2, 3, 4, 5, 6, 10, 20, 35\}$.

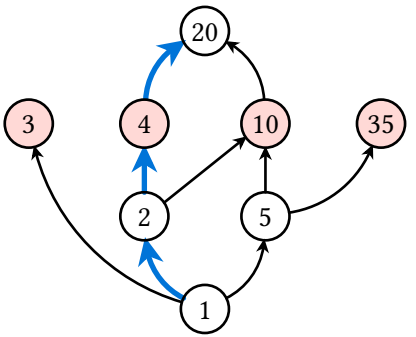
Chains: (totally ordered subsets)

- Maximal chain: $\{1, 2, 4, 12\}$ — longest path
- Not maximal: $\{5, 10\}$
- Can *skip* elements: $\{1, 5, 20\}$

Maximal elements: 3, 20, 35

Antichains: (pairwise incomparable elements)

- Maximal antichain: $\{3, 4, 10, 35\}$
- Incomparable on the same level: $\{2, 5\}$
- Incomparable from different levels: $\{3, 2, 35\}$



Dilworth's theorem: Maximum antichain size (4) = minimum chains needed to cover (4).

Examples of Chains and Antichains [2]

Example: In a Git repository, commits form a poset under the "ancestor" relation:

- **Chain:** A sequence of commits on a single branch (linear history).
- **Antichain:** Commits on different branches that have diverged (no ancestry relation).

Practical insight: Merge commits combine multiple antichains back into a single chain.

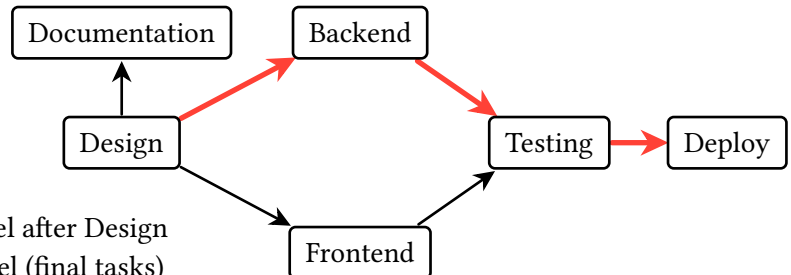
Chains and Antichains in Scheduling

Example: In project management, tasks form a scheduling poset under the "*prerequisite*" relation.

Consider web development tasks: Design, Backend, Frontend, Testing, Deploy, Documentation.

Dependencies:

- Design $\prec$ Back, Front
- Back, Front $\prec$ Test
- Test $\prec$ Deploy $\prec$ Doc



Antichain analysis:

- {Back, Front} can run in parallel after Design
- {Deploy, Doc} can run in parallel (final tasks)

Critical path: Design $\rightarrow$ Back $\rightarrow$ Test $\rightarrow$ Deploy (length 3 max chain)

Practical insights:

- Chains = sequential dependencies (critical path)
- Antichains = tasks for parallel execution (resource allocation)
- Project duration = length of longest chain

Dilworth's Theorem

Theorem 3 (Dilworth): In any finite partially ordered set, the maximum size of an antichain equals the minimum number of chains needed to cover the entire set.

Proof: Let $(P, \leq)$ be a finite poset. Let α denote the maximum size of an antichain in P, and let β denote the minimum number of chains needed to cover P. We prove $\alpha = \beta$ by showing $\alpha \leq \beta$ and $\alpha \geq \beta$.

Part 1 ($\alpha \leq \beta$): Suppose P can be covered by k chains $C_1, C_2, \ldots, C_k$. Let A be any antichain in P.

Since elements in an antichain are pairwise incomparable, no two elements of A can belong to the same chain. Therefore, A can contain at most one element from each chain C_i.

Thus $|A| \leq k$, and taking the maximum over all antichains: $\alpha \leq k$.

Since this holds for any chain decomposition with k chains: $\alpha \leq \beta$.

Dilworth's Theorem [2]

Part 2 ($\alpha \geq \beta$): We construct a chain decomposition using exactly α chains.

Define the *height* of element $x \in P$ as: $h(x) = \max\{|C| : C \text{ is a chain with maximum element } x\}$

Claim: For each $i = 1, 2, ..., \alpha$, the set $L_i = \{x \in P : h(x) = i\}$ is an antichain.

Proof of Claim: Suppose $x, y \in L_i$ with $x < y$. Then any maximal chain ending at x can be extended by adding y, so $h(y) \geq h(x) + 1 = i + 1$, contradicting $h(y) = i$.

Since P is finite, $h(x) \leq \alpha$ for all $x \in P$. The sets $L_1, ..., L_\alpha$ partition P into antichains.

For each $x \in L_i$ with $i > 1$, there exists $y < x$ with $h(y) = i - 1$ (by maximality of chains). This defines chains that cover all of P using exactly α chains.

Therefore $\beta \leq \alpha$, completing the proof. $\square$

Functions

"A function is a machine which converts a certain class of inputs into a certain class of outputs."

— Norbert Wiener

Leonhard Euler

Augustin-Louis Cauchy

Karl Weierstrass

Joseph-Louis Lagrange

George Pólya

Norbert Wiener

Definition of a Function

Definition 35: A *function* f from a set A to a set B, denoted $f : A \to B$, is a special kind of relation $f \subseteq A \times B$ where every element of A is paired with *exactly one* element of B.

This means two conditions must hold:

1. *Functional (right-unique)*: For every $a \in A$, there is *at most one* pair $\langle a, b \rangle$ in f.

$$\forall a \in A. \forall b_1, b_2 \in B. (f(a) = b_1) \wedge (f(a) = b_2) \to (b_1 = b_2)$$

2. *Serial (left-total)*: For every $a \in A$, there is *at least one* pair $\langle a, b \rangle$ in f.

$$\forall a \in A. \exists b \in B. f(a) = b$$

Definition 36: A relation that satisfies the *functional* property is called a *partial function*.

A relation that satisfies *both* properties is called a *total function*, or simply a *function*.

Domain, Codomain, Range

Definition 37: For a function $f : A \to B$:

- The set A is called the *domain* of f, denoted $\mathrm{Dom}(f)$.
- The set B is called the *codomain* of f, denoted $\mathrm{Cod}(f)$.
- The *range* (or *image*) of f is the set of all values that f actually takes:

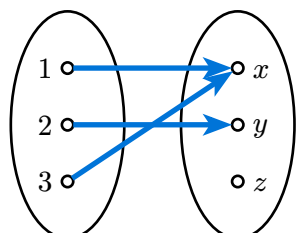
$$\mathrm{Range}(f) = \{b \in B \mid \exists a \in A.\, f(a) = b\} = \{f(a) \mid a \in A\}$$

Note: $\mathrm{Range}(f) \subseteq \mathrm{Cod}(f)$

Example: Let $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$. Let $f = \{\langle 1, x\rangle, \langle 2, y\rangle, \langle 3, x\rangle\}$.

- f is a function from A to B.
- $\mathrm{Dom}(f) = A$
- $\mathrm{Cod}(f) = B$
- $\mathrm{Range}(f) = \{x, y\} \subseteq B$

We have $f(1) = x$, $f(2) = y$, $f(3) = x$.



Domain, Codomain, Range [2]

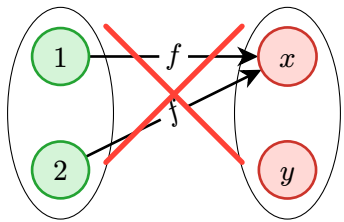
Example: Consider $g : \mathbb{Z} \to \mathbb{Z}$ defined by $g(n) = n^2$.

- $\mathrm{Dom}(g) = \mathbb{Z}$.
- $\mathrm{Cod}(g) = \mathbb{Z}$.
- $\mathrm{Range}(g) = \{0, 1, 4, 9, \ldots\}$ (the set of non-negative perfect squares).

Injective Functions

Definition 38: A function $f : A \to B$ is *injective* (or *one-to-one*[3]) if distinct elements in the domain map to distinct elements in the codomain. Formally:

$$\forall a_1, a_2 \in A. \, (f(a_1) = f(a_2)) \to (a_1 = a_2)$$



Example: $f : \mathbb{N} \to \mathbb{N}$ defined by $f(n) = 2n$ is injective. If $f(n_1) = f(n_2)$, then $2n_1 = 2n_2$, so $n_1 = n_2$.

Example: $g : \mathbb{Z} \to \mathbb{Z}$ defined by $g(n) = n^2$ is *not* injective, because $g(-1) = 1$ and $g(1) = 1$, but $-1 \neq 1$.

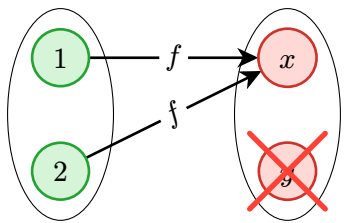
[3] Do not confuse it with *one-to-one correspondence*, which is a bijection, not just injection!

Surjective Functions

Definition 39: A function $f : A \to B$ is *surjective* (or *onto*) if every element in the codomain is the image of at least one element in the domain. Formally:

$$\forall b \in B. \exists a \in A. f(a) = b$$

For surjective functions, $\text{Range}(f) = \text{Cod}(f)$, *i.e.*, there are *no "uncovered"* elements in the right side.



Example: $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$ is surjective. For any $y \in \mathbb{R}$, $x = \sqrt[3]{y}$ is such that $f(x) = y$.

Example: $g : \mathbb{N} \to \mathbb{N}$ defined by $g(n) = 2n$ is *not* surjective, because odd numbers in $\mathbb{N}$ (the codomain) are not in the range of g. For example, there is no $n \in \mathbb{N}$ such that $2n = 3$.

Bijective Functions

Definition 40: A function $f : A \to B$ is *bijective* if it is both injective and surjective. A bijective function establishes a *one-to-one correspondence* between the elements of A and B.

Example: $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x + 1$ is bijective.

- Injective: If $2x_1 + 1 = 2x_2 + 1$, then $2x_1 = 2x_2$, so $x_1 = x_2$.
- Surjective: For any $y \in \mathbb{R}$, let $x = \frac{y-1}{2}$. Then $f(x) = 2\left(\frac{y-1}{2}\right) + 1 = y - 1 + 1 = y$.

Example: The identity function $\mathrm{id}_A : A \to A$ defined by $\mathrm{id}_A(x) = x$ for all $x \in A$ is bijective.

Function Composition

Definition 41: Let $f : A \to B$ and $g : B \to C$ be two functions. The *composition* of g and f, denoted $g \circ f$ (read as "g composed with f" or "g after f"), is a function from A to C defined by:

$$(g \circ f)(a) = g(f(a))$$

Example: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$ and $g : \mathbb{R} \to \mathbb{R}$ be $g(x) = x + 1$.

- $(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1$.
- $(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2 = x^2 + 2x + 1$.

Properties of Function Composition

- *Associativity:* If $f : A \to B$, $g : B \to C$, and $h : C \to D$, then $(h \circ g) \circ f = h \circ (g \circ f)$.
- The *identity* function acts as a *neutral* element for composition:
  - $\mathrm{id}_B \circ f = f$ for any function $f : A \to B$.
  - $f \circ \mathrm{id}_A = f$ for any function $f : A \to B$.
- Composition *preserves* the properties of functions:
  - If f and g are injective, so is $g \circ f$.
  - If f and g are surjective, so is $g \circ f$.
  - If f and g are bijective, so is $g \circ f$.
- Note that in general, $g \circ f \neq f \circ g$, *i.e.*, function composition is *not commutative*.

Inverse Functions

Definition 42: If $f : A \to B$ is a bijective function, then its *inverse function*, denoted $f^{-1} : B \to A$, is defined as:

$$f^{-1}(b) = a \quad \text{iff} \quad f(a) = b$$

Note: A function has an inverse *if and only if* it is bijective.

Example: Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 2x + 1$. We found it's bijective. To find $f^{-1}(y)$, let $y = 2x + 1$. Solving for x, we get $x = \frac{y-1}{2}$. So, $f^{-1}(y) = \frac{y-1}{2}$.

Theorem 4: If $f : A \to B$ is a bijective function with inverse $f^{-1} : B \to A$:

- f^{-1} is also bijective.
- $(f^{-1} \circ f)(a) = a$ for all $a \in A$ (*i.e.*, $f^{-1} \circ f = \mathrm{id}_A$).
- $(f \circ f^{-1})(b) = b$ for all $b \in B$ (*i.e.*, $f \circ f^{-1} = \mathrm{id}_B$).
- If $f : A \to B$ and $g : B \to C$ are both bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Image and Preimage of Sets

Definition 43: Let $f : A \to B$ be a function and let $S \subseteq A$. The *image of S under f* is the set:

$$f(S) = \{f(s) \mid s \in S\}$$

Note that $f(S) \subseteq B$. The range of f is $f(A)$.

Definition 44: Let $f : A \to B$ be a function and let $T \subseteq B$. The *preimage of T under f* (or *inverse image of T*) is the set of all elements in the domain that map into T:

$$f^{-1}(T) = \{a \in A \mid f(a) \in T\}$$

Note: The notation $f^{-1}(T)$ is used even if the inverse function f^{-1} does not exist (*i.e.*, if f is not bijective). It always refers to the set of domain elements that map into T.

Image and Preimage of Sets [2]

Example: Let $f : \mathbb{Z} \to \mathbb{Z}$ be $f(x) = x^2$.

- Let $S = \{-2, -1, 0, 1, 2\}$. Then $f(S) = \{f(-2), f(-1), f(0), f(1), f(2)\} = \{4, 1, 0, 1, 4\} = \{0, 1, 4\}$.
- Let $T_1 = \{1, 9\}$. The preimage is $f^{-1}(T_1) = \{x \in \mathbb{Z} \mid x^2 \in \{1, 9\}\} = \{-3, -1, 1, 3\}$.
- Let $T_2 = \{2, 3\}$. The preimage is $f^{-1}(T_2) = \{x \in \mathbb{Z} \mid x^2 \in \{2, 3\}\} = \emptyset$.

Cardinality & Infinity

"God made the integers, all else is the work of man."

— Leopold Kronecker

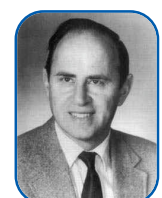
Giuseppe Peano

Leopold Kronecker

David Hilbert

Kurt Gödel

John von Neumann



Paul Cohen

Size of Sets

Definition 45: The *size* of a *finite* set X, denoted $|X|$, is the number of elements it contains.

Examples:

- Let $A = \{$🪐, 🦕, 🎻$\}$, then $|A| = 3$, since A contains *exactly 3* elements.
- Let $B = \{$🥝, 🥝, 🥝$\}$, then $|B| = 1$, since B contains *only one unique* element (the kiwi).
- $|\mathcal{P}(\{1, 2,$ 🐈$\})| = 2^3 = 8$, since the power set consists of *all 8 possible subsets* of $\{1, 2,$ 🐈$\}$.
- $|\emptyset| = 0$, since the *empty* set contains *no elements*.
- $|\mathbb{N}| = \infty$, since there are *infinitely many* natural numbers.
- $|\mathbb{R}| = \infty$, since there are *infinitely many* real numbers.

Cardinality of Sets

Definition 46: The *cardinality* of a set X, denoted $|X|$, is a measure of its "size".

- For *finite* sets, cardinality $|X|$ is the same as size, *i.e.*, the number of elements in X.
- For *infinite* sets, cardinality $|X|$ describes the "type" of infinity, *e.g.* *countable* vs *uncountable*.

Examples:

- $|\mathbb{N}| = \aleph_0$
- $|\mathbb{Q}| = \aleph_0$
- $|\mathbb{R}| = 2^{\aleph_0} = \mathfrak{c}$

Note: $|X|$ is *not* just a number, but a *cardinal number*.

- Cardinal numbers extend natural numbers to describe sizes of infinite sets.
- The *finite* cardinal numbers are just natural numbers: $0, 1, 2, 3, \ldots$.
- The first (smallest) *infinite* cardinal is $\aleph_0$ (the cardinality of $\mathbb{N}$).
- *Arithmetic* operations on cardinal numbers *differ* from those on natural numbers.
  - For example, $\aleph_0 + 1 = \aleph_0$ and $\aleph_0 \cdot 2 = \aleph_0$.

Equinumerosity

Definition 47: Two sets A and B have the same *cardinality* and called *equinumerous*, denoted $|A| = |B|$ or $A \approx B$, iff there is a *bijection* (one-to-one correspondence) from A to B.

Theorem 5: Equinumerosity is an equivalence relation.

Proof: Let A, B, C be sets.

- *Reflexivity:* The identity map $\mathrm{id}_A : A \to A$, where $\mathrm{id}_A(x) = x$, is a bijection, so $A \approx A$.
- *Symmetry:* Suppose $A \approx B$, then there is a bijection $f : A \to B$. Since it is a bijection, its inverse f^{-1} exists and is also a bijection. Hence, $f^{-1} : B \to A$ is a bijection, so $B \approx A$.
- *Transitivity:* Suppose that $A \approx B$ and $B \approx C$, *i.e.*, there are bijections $f : A \to B$ and $g : B \to C$. Then the composition $g \circ f : A \to C$ is also a bijection. So $A \approx C$.

$\square$

Countable Sets

Definition 48: A set called *countable* if it is either finite or has the same cardinality as the set of natural numbers $\mathbb{N}$. Alternatively, a set is countable if there is a *bijection* from $\mathbb{N}$ to that set.

When an infinite set is *countable*, its cardinality is denoted $\aleph_0$ (*"aleph-null"*).

Example: $|\mathbb{N}_{\text{odd}} = \{1, 3, 5, \ldots\}| = \aleph_0$, the set of *odd* natural numbers is countable, since there is a bijection $f : \mathbb{N} \to \mathbb{N}_{\text{odd}}$ defined by $f(n) = 2n + 1$.

Example: $|\{x \in \mathbb{N} \mid x \text{ is prime}\}| = \aleph_0$, the set of *prime* numbers is countable.

Example: $|\mathbb{Z}| = \aleph_0$, the set of *integers* $(-\infty, \ldots, -2, -1, 0, 1, 2, \ldots, \infty)$ is countable, since there is a bijection $f : \mathbb{N} \to \mathbb{Z}$ defined by $f(n)$:

$$f(n) = (-1)^n \left\lceil \frac{n}{2} \right\rceil = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{n+1}{2} & \text{if } n \text{ is odd} \end{cases} \qquad \begin{bmatrix} f(0) & f(1) & f(2) & f(3) & f(4) & f(5) & f(6) & \ldots \\ \left\lceil \frac{0}{2} \right\rceil & -\left\lceil \frac{1}{2} \right\rceil & \left\lceil \frac{2}{2} \right\rceil & -\left\lceil \frac{3}{2} \right\rceil & \left\lceil \frac{4}{2} \right\rceil & -\left\lceil \frac{5}{2} \right\rceil & \left\lceil \frac{6}{2} \right\rceil & \ldots \\ 0 & -1 & 1 & -2 & 2 & -3 & 3 & \ldots \end{bmatrix}$$

Countability Constructions

Definition 49: A set X is *enumerable* if there is a surjection $e : \mathbb{N} \to X$ (equivalently a bijection with either $\mathbb{N}$ or an initial segment of $\mathbb{N}$ if X finite).

Theorem 6 (Zig-Zag Enumeration): $\mathbb{N}^2$ is countable.

Proof: List pairs by diagonals of constant sum: $\langle 0, 0\rangle$; $\langle 0, 1\rangle, \langle 1, 0\rangle$; $\langle 0, 2\rangle, \langle 1, 1\rangle, \langle 2, 0\rangle$; ... giving a bijection with $\mathbb{N}$. $\square$

Theorem 7: $\mathbb{Q}$ is countable.

Proof: Enumerate positive reduced fractions p/q ordered by $p + q$ and increasing p; skip non-reduced. Interleave 0 and negatives. This yields *enumeration*, hence $\mathbb{Q} \approx \mathbb{N}$. $\square$

Pairing Functions

Definition 50: A function $f : A \times B \to \mathbb{N}$ is an arithmetical *pairing function* if it is injective.

We say that f *encodes* $A \times B$, and that $f(a, b)$ is the *code* of the pair $\langle a, b \rangle$.

Example: The *Cantor pairing function* $g : \mathbb{N}^2 \to \mathbb{N}$ is defined as:

$$g(n, k) = \frac{(n + k + 1)(n + k)}{2} + n$$

Georg Cantor

Uncountable Sets

Definition 51: A set is *uncountable* if it is not countable.

In order to prove that a set A is *uncountable*, we need to show that *no bijection $\mathbb{N} \to A$ can exist*.

The general strategy for showing that is to use *Cantor's diagonal argument*. Given a list of elements of A, say $x_1, x_2, \ldots$ (enumerated by natural numbers), we construct a *new* element of A that *differs* from each x_i, thus showing that the list cannot be complete, and hence no bijection can exist.

Theorem 8: B^ω is uncountable.

Proof: Recall that $\mathbb{B}^\omega$ is the set of all *infinite sequences* of elements from $\mathbb{B} = \{0, 1\}$.
For example, $\mathbb{B}^\omega$ contains sequences like 0000..., 010101..., 1110..., *etc.*

Suppose for contradiction that $\mathbb{B}^\omega$ is countable. Then we can *enumerate* its elements as $x_1, x_2, \ldots$, where each x_i is an infinite sequence of bits, so we can represent it as $x_i = (b_{i1}, b_{i2}, b_{i3}, \ldots)$, where $b_{ij} \in \mathbb{B}$ is the j-th bit of the i-th sequence.

Uncountable Sets [2]

Now we construct a new sequence $\Delta = \left(\overline{b}_{11}, \overline{b}_{22}, \overline{b}_{33}, \ldots\right)$, where $\overline{b}_{ii} = 1 - b_{ii}$, *i.e.*, we flip the i-th bit of the i-th sequence. This sequence *differs* from each x_i at least in the i-th position, so it cannot be equal to any x_i, so it is *not in* the enumeration $x_1, x_2, \ldots$.

	1	2	3	...
x_1	$\boldsymbol{b_{11}}$	b_{12}	b_{13}	...
x_2	b_{21}	$\boldsymbol{b_{22}}$	b_{23}	...
x_3	b_{31}	b_{32}	$\boldsymbol{b_{33}}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
Δ	$\overline{b}_{11}$	$\overline{b}_{22}$	$\overline{b}_{33}$	...

Since Δ is constructed from the bits, it is also an *element* of $\mathbb{B}^\omega$. Thus, we have found an element of $\mathbb{B}^\omega$ that is not in the enumeration $x_1, x_2, \ldots$, contradicting the assumption that $\mathbb{B}^\omega$ is countable. $\square$

Sets of Different Sizes

Definition 52: The cardinality of a set A is *less than or the same* as the cardinality of a set B, denoted $|A| \leq |B|$ or $A \preceq B$, if there is an *injection* (one-to-one function) from A to B.

Definition 53: Set A is *smaller* than B, denoted $|A| < |B|$ or $A \prec B$, iff there is an *injection*, but *no bijection* from A to B, *i.e.*, $A \preceq B$ and $A \not\approx B$.

Note: Using this notation, we can say that a set X is *countable* iff $X \preceq \mathbb{N}$, and *uncountable* iff $\mathbb{N} \prec X$.

Example: $\{1, 2\} \prec \{a, b, c\}$, since there is an injection $f : \{1, 2\} \to \{a, b, c\}$ defined by $f(1) = a$ and $f(2) = b$, but no bijection exists.

Example: $\mathbb{N} \preceq \mathbb{Z}$, since there is bijection (and thus an injection) $f : \mathbb{N} \to \mathbb{Z}$.

Example: $\mathbb{Z} \preceq \mathbb{N}$, since there is bijection (and thus an injection) $f : \mathbb{Z} \to \mathbb{N}$.

Example: $\mathbb{N} \prec \mathcal{P}(\mathbb{N})$, since there is an injection $f(x) = \{x\}$, but no bijection exists.

Cantor's Theorem

Theorem 9 (Cantor): $A \prec \mathcal{P}(A)$, for any set A.

Proof: The map $f(x) = \{x\}$ is an injection $f : A \to \mathcal{P}(A)$, since if $x \neq y$, then also $\{x\} \neq \{y\}$ by extensionality, and so $f(x) \neq f(y)$. So we have that $A \preceq \mathcal{P}(A)$.

It remains to show that $A \not\approx B$. For reductio, suppose $A \approx B$, *i.e.*, there is some bijection $g : A \to B$. Now consider $D = \{x \in A \mid x \notin g(x)\}$. Note that $D \subseteq A$, so $D \in \mathcal{P}(A)$. Since g is a bijection, there exists some $y \in A$ such that $g(y) = D$. But now we have

$$y \in g(y) \text{ iff } y \in D \text{ iff } y \notin g(y)$$

This is a contradiction, since y cannot be both *in* and *not in* $g(y)$. Thus, $A \not\approx \mathcal{P}(A)$. $\square$

Schröder–Bernstein Theorem

Theorem 10 (Schröder–Bernstein): If $A \preceq B$ and $B \preceq A$, then $A \approx B$.

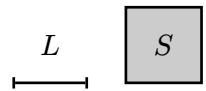
In other words, if there are injections in both directions between two sets, then there is a bijection.

Proof: Obvious, but difficult. 🤷 $\square$

Another Cantor's Theorem

Let L be the unit line, *i.e.*, the set of points $[0, 1]$. Let S be the unit square, *i.e.*, the set of points $L \times L$.

Theorem 11: $L \approx S$.



Proof[4]: Consider the function $f : L \to S$ defined by $f(x) = (x, x)$. This is an injection, since if $f(a) = f(b)$, then $(a, a) = (b, b)$, so $a = b$. Thus, $L \preceq S$.

Now consider the function $g : S \to L$ that maps (x, y) to the real number obtained by *interleaving* the decimal expansions of x and y.

$$\left.\begin{aligned} x &= 0.x_1x_2x_3\ldots \\ y &= 0.y_1y_2y_3\ldots \end{aligned}\right\} \quad g(x, y) = 0.x_1y_1x_2y_2x_3y_3\ldots$$

This is an injection, since if $g(a, b) = g(c, d)$, then $a_n = c_n$ and $b_n = d_n$ for all $n \in \mathbb{N}$, so $(a, b) = (c, d)$. Thus, $S \preceq L$.

By Schröder–Bernstein (Theorem 10), we have that $L \approx S$. $\square$

[4] See https://math.stackexchange.com/a/183383 for more detailed analysis.

Closures of Relations

Closures of Relations

Definition 54: The *closure* of a relation $R \subseteq M^2$ with respect to a property P is the smallest relation containing R that satisfies property P.

- *Reflexive closure*: $r(R) = R \cup I_M$ (smallest reflexive relation containing R)
- *Symmetric closure*: $s(R) = R \cup R^{-1}$ (smallest symmetric relation containing R)
- *Transitive closure*: $t(R)$ is the smallest transitive relation containing R

The key insight is that closure operations *add the minimum* number of pairs needed to achieve the desired property, while preserving all existing pairs in the original relation.

Reflexive Closure

Definition 55: The *reflexive closure* $r(R)$ of a relation $R \subseteq M^2$ is defined as:

$$r(R) = R \cup I_M = R \cup \{\langle x, x\rangle \mid x \in M\}$$

Example: Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle\}$.

The identity relation is $I_M = \{\langle 1, 1\rangle, \langle 2, 2\rangle, \langle 3, 3\rangle\}$.

The reflexive closure is:

$$r(R) = R \cup I_M = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 3\rangle\}$$

Symmetric Closure

Definition 56: The *symmetric closure* $s(R)$ of a relation $R \subseteq M^2$ is defined as:

$$s(R) = R \cup R^{-1} = R \cup \{\langle b, a\rangle \mid \langle a, b\rangle \in R\}$$

Example: Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle\}$.

The converse relation is:

$$R^{-1} = \{\langle 2, 1\rangle, \langle 3, 2\rangle, \langle 3, 1\rangle\}$$

The symmetric closure is:

$$s(R) = R \cup R^{-1} = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle, \langle 2, 1\rangle, \langle 3, 2\rangle, \langle 3, 1\rangle\}$$

Transitive Closure

Definition 57: The *transitive closure* $t(R)$ of a relation $R \subseteq M^2$ is the smallest transitive relation containing R.

Theorem 12: The transitive closure can be computed as:

$$t(R) = \bigcup_{n=1}^{\infty} R^n \quad \text{where } R^n = \underbrace{R \circ R \circ \ldots \circ R}_{n \text{ times}}$$

For finite sets with $|M| = k$, we have $t(R) = R^1 \cup R^2 \cup \ldots \cup R^k$.

Proof: Since M is finite, any path of length greater than $|M|$ must repeat vertices, so we only need to consider paths of length at most $|M|$. $\square$

Transitive Closure [2]

Example (Step-by-step transitive closure computation): Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$.

Step	Description	Result
Step 1:	Compute $R^1 = R$.	$R^1 = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$
Step 2:	Compute $R^2 = R \circ R$.	$R^2 = \{\langle 1, 3\rangle\}$
	For $\langle a, c\rangle \in R^2$, we need $\exists b : \langle a, b\rangle \in R \wedge \langle b, c\rangle \in R$. • $\langle 1, 3\rangle \in R^2$ since $\langle 1, 2\rangle \in R$ and $\langle 2, 3\rangle \in R$	
Step 3:	Compute $R^3 = R^2 \circ R$.	$R^3 = \emptyset$
	For $\langle a, c\rangle \in R^3$, we need $\exists b : \langle a, b\rangle \in R^2 \wedge \langle b, c\rangle \in R$. • No such pairs exist.	
Step 4:	Form the transitive closure: $t(R) = R^1 \cup R^2 \cup R^3$.	$t(R) = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle\}$

Combined Closures

Definition 58: Closure operations can be combined to create relations with multiple properties:

- *Reflexive-symmetric closure*: $rs(R) = sr(R) = r(R) \cup s(R) \cup I_M$
- *Reflexive-transitive closure*: $rt(R) = tr(R) = t(R) \cup I_M$
- *Equivalence closure*: $rst(R) = tsr(R)$ (reflexive, symmetric, and transitive)

Theorem 13 (Commutativity of closure operations):

- Reflexive and symmetric closures commute: $rs(R) = sr(R)$
- Reflexive and transitive closures commute: $rt(R) = tr(R)$
- All three closures commute when applied together

Reflexive-Symmetric Closure

Example (Reflexive-symmetric closure): Let $M = \{1, 2, 3\}$ and $R = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$.

Method 1: Apply reflexive closure first, then symmetric:

$$\begin{aligned} r(R) &= R \cup I_M \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 3\rangle\} \\ &\quad /\!/\ sr(R) = r(R) \cup r(R)^{-1} \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 3\rangle, \langle 2, 1\rangle, \langle 3, 2\rangle\} \end{aligned}$$

Method 2: Apply symmetric closure first, then reflexive:

$$\begin{aligned} s(R) &= R \cup R^{-1} \\ &= \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 2, 1\rangle, \langle 3, 2\rangle\} \\ &\quad /\!/\ rs(R) = s(R) \cup I_M \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 1\rangle, \langle 2, 2\rangle, \langle 2, 3\rangle, \langle 3, 2\rangle, \langle 3, 3\rangle\} \end{aligned}$$

Both methods yield the same result, confirming *commutativity*.

Reflexive-Transitive Closure

Example (Reflexive-transitive closure (Kleene star)): Let $M = \{a, b, c\}$ and $R = \{\langle a, b\rangle, \langle b, c\rangle\}$.

First, compute the transitive closure:

$$t(R) = R \cup R^2 = \{\langle a, b\rangle, \langle b, c\rangle, \langle a, c\rangle\}$$

Then add reflexivity:

$$rt(R) = t(R) \cup I_M = \{\langle a, a\rangle, \langle a, b\rangle, \langle a, c\rangle, \langle b, b\rangle, \langle b, c\rangle, \langle c, c\rangle\}$$

This is equivalent to the *reflexive-transitive closure*, often denoted R^* (Kleene star).

Equivalence Closure

Example: Let $M = \{1, 2, 3, 4\}$ and $R = \{\langle 1, 2\rangle, \langle 3, 4\rangle\}$.

Step 1: Make it reflexive:

$$r(R) = R \cup I_M = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 2\rangle, \langle 3, 3\rangle, \langle 3, 4\rangle, \langle 4, 4\rangle\}$$

Step 2: Make it symmetric:

$$\begin{aligned} sr(R) &= r(R) \cup r(R)^{-1} \\ &= \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 1\rangle, \langle 2, 2\rangle, \langle 3, 3\rangle, \langle 3, 4\rangle, \langle 4, 3\rangle, \langle 4, 4\rangle\} \end{aligned}$$

Step 3: Make it transitive:

- Since $\langle 1, 2\rangle, \langle 2, 1\rangle \in sr(R)$, transitivity requires $\langle 1, 1\rangle$ (already present).
- Since $\langle 3, 4\rangle, \langle 4, 3\rangle \in sr(R)$, transitivity requires $\langle 3, 3\rangle$ (already present).

$$tsr(R) = sr(R) \quad \text{(no new pairs needed)}$$

The equivalence closure partitions M into equivalence classes $\{1, 2\}$ and $\{3, 4\}$.

Equivalence Closure [2]

Example: Let $M = \{a, b, c, d, e\}$ and $R = \{\langle a, b\rangle, \langle b, c\rangle, \langle d, e\rangle\}$.

Reflexive closure:

$$r(R) = R \cup \{\langle a, a\rangle, \langle b, b\rangle, \langle c, c\rangle, \langle d, d\rangle, \langle e, e\rangle\}$$

Symmetric closure:

$$sr(R) = r(R) \cup \{\langle b, a\rangle, \langle c, b\rangle, \langle e, d\rangle\}$$

Transitive closure: We need to add pairs to ensure transitivity:

- From $\langle a, b\rangle, \langle b, c\rangle$: add $\langle a, c\rangle$
- From $\langle c, b\rangle, \langle b, a\rangle$: add $\langle c, a\rangle$

$$tsr(R) = sr(R) \cup \{\langle a, c\rangle, \langle c, a\rangle\}$$

The final equivalence relation has equivalence classes $\{a, b, c\}$ and $\{d, e\}$.

Warshall's Algorithm for Transitive Closure

Definition 59: *Warshall's algorithm* computes the transitive closure of a relation using dynamic programming with time complexity $O(n^3)$.

Given an $n \times n$ matrix $M = [\![R]\!]$ representing relation R:

```
M = matrix(R)
for k = 1 to n:
    for i = 1 to n:
        for j = 1 to n:
            M[i,j] := M[i,j] OR (M[i,k] AND M[k,j])
```

Example (Warshall's algorithm step-by-step): Let $X = \{1, 2, 3, 4\}$ and relation R with matrix $[\![R]\!]$:

$$M^{(0)} = [\![R]\!] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Warshall's Algorithm for Transitive Closure [2]

Iteration $k = 1$: Consider paths through vertex 1.

$$M^{(1)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

(Added $\langle 4, 2 \rangle$: path $4 \to 1 \to 2$)

Iteration $k = 2$: Consider paths through vertex 2.

$$M^{(2)} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

(Added $\langle 1, 3 \rangle$ and $\langle 4, 3 \rangle$)

Warshall's Algorithm for Transitive Closure [3]

Iteration $k = 3$: Consider paths through vertex 3.

$$M^{(3)} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(Added $\langle 1, 4 \rangle$, $\langle 2, 4 \rangle$, and $\langle 4, 4 \rangle$)

Iteration $k = 4$: Consider paths through vertex 4.

$$M^{(4)} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

(The relation becomes universal since there's a cycle)

Warshall's Algorithm for Transitive Closure [4]

Example (Reachability in graphs):

Consider a social network where R represents "follows" relationships:

$$R = \{\langle A, B\rangle, \langle B, C\rangle, \langle C, D\rangle, \langle A, E\rangle\}$$

Using Warshall's algorithm, we can determine *indirect influence*:

- A can influence C through B
- A can influence D through B and C
- The transitive closure shows all possible influence paths

This is crucial for analyzing *information propagation* in social networks, *dependency resolution* in software systems, and *route planning* in transportation networks.

Properties of Closures

Theorem 14: For any relation $R \subseteq M^2$:

1. *Idempotency*: $r(r(R)) = r(R)$, $s(s(R)) = s(R)$, $t(t(R)) = t(R)$
2. *Monotonicity*: If $R_1 \subseteq R_2$, then $r(R_1) \subseteq r(R_2)$, *etc.*
3. *Extensivity*: $R \subseteq r(R)$, $R \subseteq s(R)$, $R \subseteq t(R)$
4. *Distributivity over union*: $r(R_1 \cup R_2) = r(R_1) \cup r(R_2)$, *etc.*

Examples of Closures

Example (Closure of the empty relation):

Let $M = \{a, b, c\}$ and $R = \emptyset$.

- $r(\emptyset) = \emptyset \cup I_M = I_M = \{\langle a, a\rangle, \langle b, b\rangle, \langle c, c\rangle\}$
- $s(\emptyset) = \emptyset \cup \emptyset^{-1} = \emptyset$
- $t(\emptyset) = \emptyset$ (since $\emptyset^n = \emptyset$ for all $n \geq 1$)

The reflexive closure of the *empty* relation is the *identity* relation.

Examples of Closures [2]

Example (Closure of the universal relation):

Let $M = \{1, 2\}$ and $R = M \times M = \{\langle 1, 1\rangle, \langle 1, 2\rangle, \langle 2, 1\rangle, \langle 2, 2\rangle\}$.

- $r(R) = R \cup I_M = R$ (since $I_M \subseteq R$)
- $s(R) = R \cup R^{-1} = R$ (since $R = R^{-1}$ for universal relation)
- $t(R) = R$ (universal relation is already transitive)

The universal relation is its own closure under all three operations.

Examples of Closures [3]

Example (Non-commutativity with other operations):

Let $M = \{1, 2, 3\}$, $R_1 = \{\langle 1, 2\rangle\}$, and $R_2 = \{\langle 2, 3\rangle\}$.

Consider $t(R_1 \cup R_2)$ vs $t(R_1) \cup t(R_2)$:

- $R_1 \cup R_2 = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$
- $t(R_1 \cup R_2) = \{\langle 1, 2\rangle, \langle 2, 3\rangle, \langle 1, 3\rangle\}$
- $t(R_1) = \{\langle 1, 2\rangle\}$
- $t(R_2) = \{\langle 2, 3\rangle\}$
- $t(R_1) \cup t(R_2) = \{\langle 1, 2\rangle, \langle 2, 3\rangle\}$

Since $\langle 1, 3\rangle \in t(R_1 \cup R_2)$ but $\langle 1, 3\rangle \notin t(R_1) \cup t(R_2)$, we have:

$$t(R_1 \cup R_2) \neq t(R_1) \cup t(R_2)$$

However: $t(R_1) \cup t(R_2) \subseteq t(R_1 \cup R_2)$ always holds.

Examples of Closures [4]

Example (Computing equivalence classes from closure):

Let $M = \{1, 2, 3, 4, 5\}$ and $R = \{\langle 1, 3\rangle, \langle 2, 4\rangle, \langle 4, 5\rangle\}$.

The equivalence closure gives us:

$$\begin{aligned}\text{equiv}(R) &= rst(R) \\ &= \{\langle 1, 1\rangle, \langle 1, 3\rangle, \langle 2, 2\rangle, \langle 2, 4\rangle, \langle 2, 5\rangle, \langle 3, 1\rangle, \\ &\qquad \langle 3, 3\rangle, \langle 4, 2\rangle, \langle 4, 4\rangle, \langle 4, 5\rangle, \langle 5, 2\rangle, \langle 5, 4\rangle, \langle 5, 5\rangle\}\end{aligned}$$

The equivalence classes are:

- $[1]_R = \{1, 3\}$
- $[2]_R = \{2, 4, 5\}$

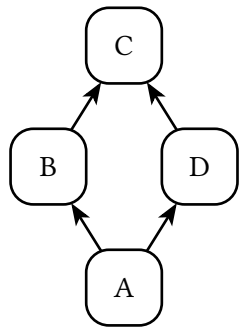
This partitions M into $\{\{1, 3\}, \{2, 4, 5\}\}$.

Examples of Closures [5]

Example: Consider a directed acyclic graph (DAG) where R represents *"depends on"* relationships:

$$R = \{\langle A, B\rangle, \langle B, C\rangle, \langle A, D\rangle, \langle D, C\rangle\}$$

For example, component A depends on B and D, which both depend on C.



The transitive closure reveals all *indirect* dependencies:

$$t(R) = R \cup \{\langle A, C\rangle\}$$

This shows that component A *transitively depends* on C through two paths:

- $A \to B \to C$
- $A \to D \to C$

In software build systems, this helps determine the complete dependency tree.

Applications of Relation Closures

Example ***(Graph Reachability)***: In directed graphs, transitive closure determines reachability:

- **Social networks:** Finding mutual connections and influence propagation
- **Transportation:** Computing all possible routes between cities
- **Citation networks:** Tracking intellectual dependencies in research
- **Web crawling:** Discovering linked pages and site connectivity

Example ***(Program Analysis)***: Closure operations enable sophisticated program optimization:

- **Data flow analysis:** Transitive closure computes variable dependencies
- **Call graph construction:** Finding all possible function invocations
- **Alias analysis:** Determining which pointers may reference the same memory
- **Taint analysis:** Tracking information flow and potential vulnerabilities

Example ***(Database Systems)***: Relational databases extensively use closure computations:

- **Recursive queries:** Computing organizational hierarchies and bill-of-materials
- **Join optimization:** Finding efficient query execution plans
- **Integrity constraints:** Ensuring referential consistency across tables
- **Data lineage:** Tracking data provenance and transformation chains

Complexity of Closure Computations

Theorem 15: For a relation R on n elements, the time complexities are:

- **Reflexive closure:** $O(n)$ time
- **Symmetric closure:** $O(|R|)$ time
- **Transitive closure:** $O(n^3)$ time (Floyd-Warshall algorithm)

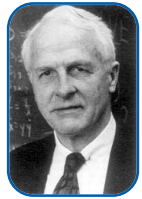
Note: The transitive closure bottleneck drives many algorithmic design decisions in practice.

Lattices

"Order is the shape upon which beauty depends."

— Pearl S. Buck

Alfred Tarski



Garrett Birkhoff

Dana Scott

Emmy Noether

Marshall Stone

Upper and Lower Bounds

Definition 60: In a poset $\langle S, \leq \rangle$, an element $u \in S$ is called an *upper bound* of a subset $C \subseteq S$ if it is greater than or equal to every element in C, *i.e.*, for all $x \in C$, $x \leq u$.

Definition 61: In a poset $\langle S, \leq \rangle$, an element $l \in S$ is called a *lower bound* of a subset $C \subseteq S$ if it is less than or equal to every element in C, *i.e.*, for all $x \in C$, $l \leq x$.

Example: In $\langle \mathbb{R}, \leq \rangle$ for interval $C = (0; 1)$:

- **Lower bounds:** every $x \leq 0$ (including $-\infty, -1, 0$)
- **Upper bounds:** every $x \geq 1$ (including $1, 2, +\infty$)
- **No** greatest lower bound or least upper bound *in* C (since $(0; 1)$ is open)

Examples of Bounds

Example: In $\langle \mathcal{P}(\{1,2,3\}), \subseteq \rangle$ for $C = \{\{1,2\},\{1,3\}\}$:

- **Lower bounds:** $\emptyset$, $\{1\}$ (subsets of both sets in C)
- **Upper bounds:** $\{1,2,3\}$ (supersets of both sets in C)
- **Greatest lower bound:** $\{1\} = \{1,2\} \cap \{1,3\}$
- **Least upper bound:** $\{1,2,3\} = \{1,2\} \cup \{1,3\}$

Example: In divisibility poset for $C = \{4,6\}$:

- **Upper bounds:** multiples of $\mathrm{lcm}(4,6) = 12$, *i.e.*, $\{12, 24, 36, \ldots\}$
- **Lower bounds:** common divisors, *i.e.*, $\{1,2\}$
- **Least upper bound:** $12 = \mathrm{lcm}(4,6)$
- **Greatest lower bound:** $2 = \gcd(4,6)$

Suprema and Infima

Definition 62: In a poset $\langle S, \leq \rangle$, the *supremum* (or *join*) of a subset $C \subseteq S$, denoted $\sup(C)$ or $\bigvee C$, is the *least upper bound* of C, *i.e.*, an upper bound $u \in S$ s.t. for any other upper bound $v \in S$, $u \leq v$.

Note: If it exists, the least upper bound is *unique*.

Definition 63: In a poset $\langle S, \leq \rangle$, the *infimum* (or *meet*) of a subset $C \subseteq S$, denoted $\inf(C)$ or $\bigwedge C$, is the *greatest lower bound* of C, *i.e.*, a lower bound $l \in S$ s.t. for any other lower bound $m \in S$, $m \leq l$.

Note: If it exists, the greatest lower bound is *unique*.

Example: $\langle \mathbb{R}, \leq \rangle$:

- For finite subsets, $\sup(C) = \max(C)$ and $\inf(C) = \min(C)$.
- For infinite subsets: $\sup((0;1)) = 1$ and $\inf((0;1)) = 0$ (even though $0, 1 \notin (0;1)$)

Examples of Suprema and Infima

Example: $\langle \mathcal{P}(A), \subseteq \rangle$:

- **Join:** $\sup(\mathcal{C}) = \bigcup_{X \in \mathcal{C}} X$ (union of all sets)
- **Meet:** $\inf(\mathcal{C}) = \bigcap_{X \in \mathcal{C}} X$ (intersection of all sets)
- $\sup\{\{1,2\},\{2,3\},\{3,4\}\} = \{1,2,3,4\}$
- $\inf\{\{1,2,3\},\{2,3,4\},\{3,4,5\}\} = \{3\}$

Example: Divisibility on $\mathbb{N}^+$:

- **Join:** $\sup\{a,b\} = \mathrm{lcm}(a,b)$ (least common multiple)
- **Meet:** $\inf\{a,b\} = \gcd(a,b)$ (greatest common divisor)
- $\sup\{6,10\} = 30$
- $\inf\{6,10\} = 2$

Lattices

Definition 64: A poset $\langle S, \leq \rangle$ where every non-empty finite subset $C \subseteq S$ has a join (supremum) is called an *upper semilattice* (or *join-semilattice*) and denoted $\langle S, \vee \rangle$.

Definition 65: A poset $\langle S, \leq \rangle$ where every non-empty finite subset $C \subseteq S$ has a meet (infimum) is called a *lower semilattice* (or *meet-semilattice*) and denoted $\langle S, \wedge \rangle$.

Definition 66: A poset $\langle S, \leq \rangle$ that is both an upper semilattice and a lower semilattice, *i.e.*, every non-empty finite subset has both a join and a meet, is called a *lattice*, denoted $(S, \vee, \wedge)$.

Why Lattices?

Why study lattices? Whenever you have:

- Elements that can be *compared* (ordered)
- Ways to *combine* elements (join, meet)
- Consistent behavior under combination

...you likely have a lattice! This structure appears in programming languages, databases, security systems, logic circuits, and many other areas of computer science and mathematics.

Properties of Lattices

Definition 67: A lattice is *bounded* if it has a greatest element $\top$ and a least element $\bot$.

Definition 68: A lattice is *distributive* if $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ (and dually).

Definition 69: A lattice is *modular* if $x \leq z$ implies $x \vee (y \wedge z) = (x \vee y) \wedge z$.

Note: Distributive $\Rightarrow$ modular.

Example (Powerset Lattice): $\langle \mathcal{P}(A), \subseteq \rangle$ is a bounded distributive lattice with $\vee = \cup$, $\wedge = \cap$, $\top = A$, $\bot = \emptyset$.

Why this matters: This is the foundation of set-based reasoning in:

- Database theory (relational algebra)
- Formal specification languages (Z, B-method)
- Model checking and verification

Examples of Lattices

Example (Divisibility Lattice): For positive integers, $a \leq b$ iff a divides b.

- **Join (LCM):** $6 \vee 10 = \mathrm{lcm}(6, 10) = 30$
- **Meet (GCD):** $6 \wedge 10 = \gcd(6, 10) = 2$
- **Bottom element:** 1 (divides everything)
- **No top element:** No integer is divisible by all others

Applications:

- Number theory and cryptography (RSA key generation)
- Computer algebra systems (polynomial GCD algorithms)
- Scheduling problems (finding common time periods)

Examples of Lattices [2]

Example (File System Permissions): Unix file permissions form a lattice under inclusion:

- Elements: Sets of permissions like $\{r, w, x\}$, $\{r, x\}$, $\{w\}$, *etc.*
- Order: $P_1 \leq P_2$ if $P_1 \subseteq P_2$ (fewer permissions $\Rightarrow$ more restrictive)
- Join: Union of permissions (less restrictive)
  - For example: $\{\text{read}\} \vee \{\text{execute}\} = \{\text{read}, \text{execute}\}$
- Meet: Intersection of permissions (more restrictive)
  - For example: $\{\text{read}, \text{write}\} \wedge \{\text{write}, \text{execute}\} = \{\text{write}\}$

Examples of Lattices [3]

Example (Partition Lattice): All partitions of a set S, ordered by refinement.

- $\pi_1 \leq \pi_2$ if π_1 is a refinement of π_2 (smaller blocks)
- Join: Finest common coarsening
  - For example: $(12 \mid 3) \vee (1 \mid 23) = (123)$
- Meet: Coarsest common refinement
  - For example: $(12 \mid 3) \wedge (1 \mid 23) = (1 \mid 2 \mid 3)$
- Applications: Clustering, database normalization

Lattices aren't just abstract algebra — they appear everywhere in computer science and mathematics.

The *join* and *meet* operations capture fundamental patterns of *combination* and *interaction*.

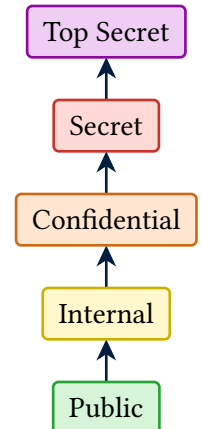
Why Lattices Matter: Information Security Levels

Example: In computer security, information has *classification levels* forming a lattice:

- Elements: $\{\text{Public}, \text{Internal}, \text{Confidential}, \text{Secret}, \text{Top Secret}\}$
- Order: $\text{Public} \leq \text{Internal} \leq \text{Confidential} \leq \text{Secret} \leq \text{Top Secret}$
- Join ($\vee$): Higher classification needed to combine information
- Meet ($\wedge$): Lower classification that both pieces can be declassified to

For instance:

- $\text{Internal} \vee \text{Confidential} = \text{Confidential}$ (combination needs higher level)
- $\text{Secret} \wedge \text{Confidential} = \text{Confidential}$ (both can be declassified to this level)



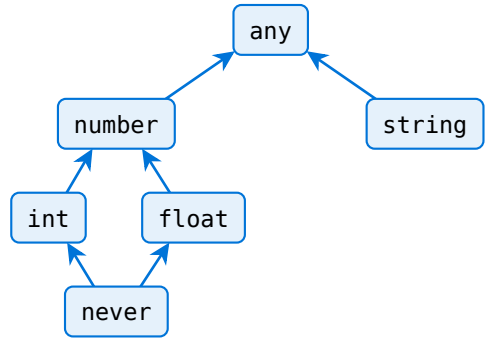
Why Lattices Matter: Type Systems

In programming language theory, *types* form lattices.

```
let x: int = 42;
let y: string = "hello";
let z = condition ? x : y;
// z has type: int ∨ string = any
```

Subtyping Lattice:

- Order: `int` $\sqsubseteq$ `number` $\sqsubseteq$ `any`, `string` $\sqsubseteq$ `any`
- Join ($\vee$): Most general common supertype (union type)
  - For example: `int` $\vee$ `string` $=$ `any`
- Meet ($\wedge$): Most specific common subtype (intersection type)
  - Let $A =$ `{ name: string }`
  - Let $B =$ `{ age: int }`
  - Then $A \wedge B =$ `{ name: string, age: int }`



Type Lattice

- Arrows show subtype relation $\sqsubseteq$
- Join: Move up to common parent
- Meet: Move down to common child

Why Lattices Matter: Program Analysis

TODO

Control Flow Analysis:

- Elements: Sets of possible program states
- Order: Subset inclusion ($\subseteq$)
- Join: Union of possible states (at merge points)
- Meet: Intersection of guaranteed properties

Why Lattices Matter: Database Query Optimization

Example: *Query execution plans* form a lattice:

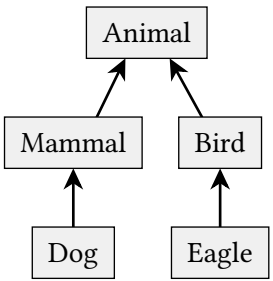
- Elements: Different ways to execute a query
- Order: "Plan A $\leq$ Plan B" if A is more efficient than B
- Join: Combine optimization strategies
- Meet: Find common optimizations

This structure helps database optimizers systematically explore the space of possible query plans.

Why Lattices Matter: Concept Hierarchies and Ontologies

Example: Knowledge representation uses *concept lattices*.

For example, consider a biological taxonomy:



- Elements: Biological concepts (*e.g.*, Animal, Mammal, Dog)
- Order: "Concept A $\leq$ Concept B" if A is a more specific type of B, *e.g.*, "Dog $\leq$ Mammal"
- Join: Most specific common ancestor, *e.g.*, "Mammal $\vee$ Bird $=$ Animal"
- Meet: Most general common descendant, *e.g.*, "Bird $\wedge$ Eagle $=$ Eagle"

Why Lattices Matter: Distributed Systems and Causality

Example: In distributed systems, *events* form a lattice *under causality*:

- Elements: System events with vector timestamps
- Order: "Event A $\leq$ Event B" if A causally precedes B
- Join: Latest information from both events
- Meet: Common causal history

This structure is crucial for:

- Consistent distributed databases
- Version control systems (Git DAG)
- Blockchain consensus algorithms

Why Lattices Matter: Logic and Boolean Reasoning

Example: *Propositional formulas* form lattices:

- Elements: Boolean formulas over variables
- Order: $\varphi \leq \psi$ if φ implies ψ (semantic entailment)
- Join: Disjunction ($\vee$) — weaker condition
- Meet: Conjunction ($\wedge$) — stronger condition

Special case: *Boolean algebra* (`true`, `false`, $\vee$, $\wedge$, $\neg$) used in:

- Digital circuit design
- Database query languages (SQL WHERE clauses)
- Search engines (Boolean search)

Well Orders

Well-Ordered Sets

Definition 70: A poset $(M, \leq)$ is *well-ordered* if every non-empty subset $S \subseteq M$ has a *least element*.

Formally: $\forall S \subseteq M. (S \neq \emptyset) \to (\exists m \in S. \forall x \in S. m \leq x)$

Note: A well-ordered set is automatically a *total order* (linear order) since comparability follows from the well-ordering property.

Example: The natural numbers $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ with the usual order $\leq$ form a well-ordered set:

- Any non-empty subset has a smallest element.
- For instance, the subset $\{5, 17, 23, 100\}$ has least element 5.
- Even infinite subsets like $\{2, 4, 6, 8, \ldots\}$ (even numbers) have a least element (2).

Example: The integers $\mathbb{Z}$ with the usual order are *not* well-ordered:

- The subset $\{-1, -2, -3, \ldots\}$ (negative integers) has no least element.
- Any infinite descending sequence has no minimum.

Examples of Well-Ordered Sets

Example: *Lexicographic order* on finite strings over an alphabet is well-ordered:

- Given any non-empty set of strings, there is always a lexicographically smallest one.
- For example, in $\{\text{cat}, \text{dog}, \text{apple}, \text{zebra}\}$, the least element is "apple".

Well-Ordered Induction

TODO

Well-Founded Relations

Definition 71: A relation $R \subseteq M^2$ is *well-founded* if every non-empty subset $S \subseteq M$ has at least one *minimal element* with respect to R.

Formally: $\forall S \subseteq M.\, (S \neq \emptyset) \to (\exists m \in S.\, \forall x \in S.\, x \not R\, m)$

Note: *Well-founded* $\neq$ *well-ordered*:

- Well-ordered requires a *least* element (unique minimum)
- Well-founded only requires *minimal* elements (no element below them)
- Every well-ordered set is well-founded, but not vice versa

Examples of Well-Founded Relations

Example: Consider the *proper subset* relation $\subset$ on finite sets. Let $M = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

Well-founded: ✓ Every subset of M has minimal elements.

- *Example:* The subset $\{\{a\}, \{b\}, \{a, b\}\}$ has minimal elements $\{a\}$ and $\{b\}$
  - Neither $\{a\} \subset \{b\}$ nor $\{b\} \subset \{a\}$ (they're incomparable)
  - Both are minimal since no set in the subset is a proper subset of them

Well-ordered: ✗ Some subsets lack a unique least element.

- *Same example:* $\{\{a\}, \{b\}, \{a, b\}\}$ has no "$\subset$-least" element
  - For "$\subset$-least", we'd need a set L such that $L \subset X$ for all other X
  - But $\{a\} \not\subset \{b\}$ and $\{b\} \not\subset \{a\}$, so neither can be least
  - No single set is a proper subset of all others in this collection

Key insight: Well-founded $\neq$ well-ordered:

- *Multiple minimal* elements are allowed in well-founded relations.
- Well-ordered relations require a *unique* least element in every subset.

Examples of Well-Founded Relations [2]

Example: The *divisibility* relation $(\mathbb{N}^+, |)$ is well-founded:

- Every non-empty subset has minimal elements (numbers that divide no others in the subset)
- For example, in $\{6, 12, 18, 4, 8\}$:
  - 4 is minimal because no other number in the set divides 4
  - 6 is minimal because no other number in the set divides 6
  - Note: 6 divides both 12 and 18, but that doesn't affect minimality
- In $\{2, 4, 8, 16\}$: only 2 is minimal (it divides all others, but nothing else divides it)

Examples of Well-Founded Relations [3]

Example: Consider the relation *"properly contains"* ($\supset$) on finite sets.

Let $S = \{\{1\}, \{2\}, \{1, 2\}, \{1, 2, 3\}\}$.

Well-founded: ✓ Every subset has minimal elements.

- The subset $\{\{1\}, \{2\}, \{1, 2, 3\}\}$ has minimal elements $\{1\}$ and $\{2\}$
  - Neither $\{1\} \supset \{2\}$ nor $\{2\} \supset \{1\}$ (they're incomparable)
  - $\{1, 2, 3\} \supset \{1\}$ and $\{1, 2, 3\} \supset \{2\}$, but they remain minimal in this subset
- Any collection always has sets that contain no others in that collection

Examples of Well-Founded Relations [4]

Example: *Program termination analysis* uses well-founded relations:

- Define a measure that decreases with each recursive call
- If the measure forms a well-founded order, the program terminates
- Example: factorial function decreases argument from n to $n - 1$

Well-Founded Induction

TODO

Induced Strict Order

Definition 72: Given a partial order $\leq$, the *induced strict order* $<$ is defined as:

$$x < y \text{ iff } (x \leq y \text{ and } x \neq y)$$

Note: Given a poset $(S, \leq)$, we can use $<$ to denote its associated strict order.

Note: $>$ is the converse of $<$, so we can write $b > a$ instead of $a < b$.

Note: *Hereinafter*, we will freely use $<$ and $>$ when given any poset $(S, \leq)$.

Descending Chain Condition

Definition 73: A poset $\langle S, \leq \rangle$ is said to satisfy the *descending chain condition (DCC)* if no strict descending sequence $x_1 > x_2 > x_3 > \ldots$ of elements of S exists.

Equivalently, every weakly descending sequence $x_1 \geq x_2 \geq x_3 \geq \ldots$ eventually stabilizes.

Formally: $\forall (x_i)_{i \in \mathbb{N}} \in S^{\mathbb{N}}. \left(\forall i \in \mathbb{N}. x_i \geq x_{i+1}\right) \to \left(\exists N \in \mathbb{N}. \forall n \geq N. x_n = x_{n+1}\right)$

Note: The term "*sequence*" does not mean we must list all elements consequently or uniquely. It simply means we have a function from $\mathbb{N}$ to S. Elements can repeat, and we can skip elements in S. However, when we say "*descending sequence*," we mean that each new element is less than the previous one.

Example: The natural numbers $(\mathbb{N}, \leq)$ satisfy DCC:

- Since natural numbers are bounded below by 0, infinite descent is impossible.
- Any (weakly) descending sequence $n_1 \geq n_2 \geq n_3 \geq \ldots$ must stabilize.
- Eventually, some $n_k = n_{k+1} = n_{k+2} = \ldots$

Well-Founded Posets

Definition 74: A poset $(S, \leq)$ is *well-founded* if its associated strict order $<$ is a well-founded relation.

In other words, every non-empty subset of S has $<$-minimal elements.

DCC and Well-Foundedness

Theorem 16: For any poset $(S, \leq)$, the following are *equivalent*:

1. $(S, \leq)$ is *well-founded* (every non-empty subset has $<$-minimal elements)
2. $(S, \leq)$ satisfies *DCC* (no infinite descending chains)

Note: Here, we again use "$<$" (and its converse "$>$") to denote the strict order induced by "$\leq$".

Proof $(\Longrightarrow)$: **Well-founded implies DCC**

Suppose $(S, \leq)$ is well-founded but there exists an infinite strict descending sequence $x_0 > x_1 > x_2 > \ldots$. Consider the set $T = \{x_0, x_1, x_2, \ldots\}$. Since $(S, \leq)$ is well-founded, T must have a minimal element x_k for some k. But then $x_k > x_{k+1}$, contradicting the minimality of x_k. $\square$

Proof $(\Longleftarrow)$: **DCC implies well-founded**

Suppose $(S, \leq)$ satisfies DCC. Let $T \subseteq S$ be any non-empty subset. If T had no minimal elements, then for any $x_0 \in T$, there would exist $x_1 \in T$ with $x_0 > x_1$. Continuing this process, we could construct an infinite strict descending sequence $x_0 > x_1 > x_2 > \ldots$, contradicting DCC. $\square$

Ascending Chain Condition

Definition 75: A poset $\langle S, \leq \rangle$ satisfies the *ascending chain condition (ACC)* if no strict ascending sequence $x_1 < x_2 < x_3 < \ldots$ of elements of S exists.

Equivalently, every weakly ascending sequence $x_1 \leq x_2 \leq x_3 \leq \ldots$ eventually stabilizes.

Example: $\langle \mathbb{N}, \leq \rangle$ does *not* satisfy ACC since infinite ascending chains like $2 < 3 < 5 < 7 < \ldots$ exist.

Example: Consider the poset $(\mathcal{P}(\mathbb{N}), \subseteq)$ of all subsets of natural numbers ordered by inclusion.

- **ACC fails:** ✗
  The ascending chain $\emptyset \subseteq \{1\} \subseteq \{1, 2\} \subseteq \{1, 2, 3\} \subseteq \ldots$ never stabilizes.
- **DCC fails:** ✗
  The descending chain $\mathbb{N} \supseteq (\mathbb{N} \setminus \{1\}) \supseteq (\mathbb{N} \setminus \{1, 2\}) \supseteq \ldots$ never stabilizes.

Noetherian Relations

Definition 76: A relation R is *Noetherian* (or *converse well-founded*, or *upwards well-founded*) if the converse relation R^{-1} is well-founded.

Equivalently, $R \subseteq M^2$ is Noetherian if every non-empty subset of M has R-maximal elements.

Note: This means no infinite ascending chains $x_0 \; R \; x_1 \; R \; x_2 \; R \; \ldots$ exist.

Example: The usual order $\leq$ on $\mathbb{N}$ is NOT Noetherian:

- We can construct infinite ascending chains like $1 < 2 < 3 < 4 < \ldots$
- However, its converse $\geq$ would be well-founded (DCC holds for $\leq$)

Noetherian Relations [2]

Example: In ring theory, a *Noetherian ring* has the property that every ascending chain of ideals stabilizes: $I_1 \subseteq I_2 \subseteq I_3 \subseteq \ldots$ eventually becomes constant.

Example: In rewriting systems and lambda calculus:

- A *reduction relation* $\rightarrow$ is Noetherian if all reduction sequences terminate
- For example, β-reduction in simply typed lambda calculus is Noetherian
- This guarantees that programs always terminate (no infinite loops)

Noetherian Relations and ACC

Theorem 17: For any poset $\langle S, \leq \rangle$, the following are equivalent:

- $\langle S, \leq \rangle$ is Noetherian (coverse well-founded)
- $\langle S, \leq \rangle$ satisfies ACC (no infinite ascending chains)

Proof: By definition, $(S, \leq)$ is Noetherian iff its converse $\geq$ is well-founded. By the earlier theorem, this is equivalent to $\geq$ satisfying DCC, which is exactly the same as $\leq$ satisfying ACC. $\square$

Relationships Between Chain Conditions

Theorem 18: For a poset $(S, \leq)$:

- **DCC** $\iff$ the relation $\leq$ is well-founded $\iff$ no infinite descending chains.
- **ACC** $\iff$ the *dual* relation $\geq$ is well-founded $\iff$ no infinite ascending chains.

Proof: The first equivalence (DCC $\iff$ well-founded) was proved earlier. The second equivalence for ACC follows by applying the same reasoning to the dual relation $\geq$. $\square$

Examples of ACC and DCC

Example: Consider the poset $\langle \mathcal{P}(\{1, 2, 3\}), \subseteq \rangle$ of subsets ordered by inclusion:

- **ACC holds:** ✓ Any ascending chain $A_1 \subseteq A_2 \subseteq A_3 \subseteq \ldots$ must stabilize since we can't keep adding elements indefinitely.
- **DCC holds:** ✓ Any descending chain $B_1 \supseteq B_2 \supseteq B_3 \supseteq \ldots$ must stabilize since we can't keep removing elements indefinitely.
- Both conditions hold because the set is finite.

Example: In the natural numbers $(\mathbb{N}, \leq)$:

- **DCC holds:** ✓ Any sequence $n_1 \geq n_2 \geq n_3 \geq \ldots$ must stabilize (well-founded).
- **ACC fails:** ✗ The sequence $1 < 2 < 3 < 4 < \ldots$ never stabilizes.
- This shows that DCC and ACC are independent conditions.

Examples (Applications in algebra):

- **Noetherian rings:** Satisfy ACC for ideals (every ascending chain of ideals stabilizes).
- **Artinian rings:** Satisfy DCC for ideals (every descending chain of ideals stabilizes).
- **Principal ideal domains:** Both conditions hold, enabling algorithms like Euclidean division.

Connections and Applications

Theorem 19 (Well-ordering principle): Every well-ordered set admits *transfinite induction*: to prove $P(x)$ for all $x \in S$, it suffices to show: $\forall x \in S.\, (\forall y < x.\, P(y)) \to P(x)$

Example: *Mathematical induction* on $\mathbb{N}$ is a special case of transfinite induction using the well-ordering of natural numbers.

Examples (Computer science applications):

- **Termination analysis:** Prove programs terminate by finding well-founded measures.
- **Parsing algorithms:** Use well-founded recursion on parse tree depth.
- **Datalog evaluation:** Stratified negation ensures termination via well-founded semantics.
- **Model checking:** Well-founded relations ensure finite state exploration.

These concepts provide the mathematical foundation for reasoning about *termination*, *finiteness*, and *algorithmic complexity* in computer science and mathematics.

Summary: Well-Founded Relations and Chain Conditions

Concept	Characterization	Key Property
Well-ordered	Every subset has unique least element	Stronger than well-founded
Well-founded	Every subset has minimal elements	DCC: no infinite descent
DCC (Artinian)	No infinite descending chains	Same as well-founded
ACC (Noetherian)	No infinite ascending chains	Dual of DCC

Key relationships:

- Well-ordered $\Longrightarrow$ Total order (least elements imply comparability)
- Well-ordered $\Longrightarrow$ Well-founded (every least element is minimal)
- Well-founded $\Leftrightarrow$ Noetherian $\Leftrightarrow$ DCC (equivalent characterizations)
- ACC is independent of DCC (a structure can satisfy one but not the other)
- These concepts are fundamental for proving termination and finiteness properties

Outline

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TODO

- Applications of lattices in:
  - Formal concept analysis
  - Domain theory in computer science
  - Algebraic topology
  - Cryptography (lattice-based cryptography)
- Advanced topics in set theory:
  - Cardinal arithmetic
  - Ordinal numbers
  - Forcing and independence results
  - Large cardinals
- Connections to Boolean algebra (next lecture)
- Applications in formal logic and proof theory

Looking Ahead: Boolean Algebra

The next lecture will explore *Boolean algebra*, which provides the mathematical foundation for:

- Digital circuit design and computer hardware
- Propositional logic and automated reasoning
- Database query optimization
- Formal verification of software and hardware systems

Key topics will include:

- Boolean functions and their representations
- Normal forms (CNF, DNF)
- Minimization techniques (Karnaugh maps, Quine-McCluskey)
- Functional completeness and Post's theorem
- The satisfiability problem (SAT) and its computational complexity

Preview: Formal Logic

Following Boolean algebra, we will study *formal logic*, covering:

- Propositional and predicate logic
- Natural deduction and proof systems
- Completeness and soundness theorems
- Applications to program verification and AI reasoning

This progression from sets $\rightarrow$ relations $\rightarrow$ functions $\rightarrow$ Boolean algebra $\rightarrow$ logic provides a solid foundation for advanced topics in discrete mathematics and computer science.

Binary relations are the bridge between sets and functions — they model how objects *connect*, *organize*, and *interact* in mathematical structures and real-world systems.